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Park et al.

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(54) APPARATUS FOR MANUFACTURING DISPLAY DEVICE

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(22) Filed: **Jun. 26, 2024**

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(30) **Foreign Application Priority Data**

Nov. 30, 2023 (KR) 10-2023-0170769

(51) **Int. Cl.**

G02B 27/14	(2006.01)
B23K 26/352	(2014.01)
B23K 26/70	(2014.01)
B23K 103/00	(2006.01)
H01L 21/02	(2006.01)
H10K 71/40	(2023.01)

(52) **U.S. Cl.**

CPC ***G02B 27/141*** (2013.01); ***B23K 26/352*** (2015.10); ***B23K 26/704*** (2015.10); ***B23K 26/705*** (2015.10); *B23K 2103/56* (2018.08); *H01L 21/02686* (2013.01); *H10K 71/421* (2023.02)

(58) **Field of Classification Search**

CPC .. G02B 27/141; B23K 26/352; B23K 26/705; B23K 2103/56; H10K 71/421; H01L 21/02686

See application file for complete search history.

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Assistant Examiner — Don J Williams
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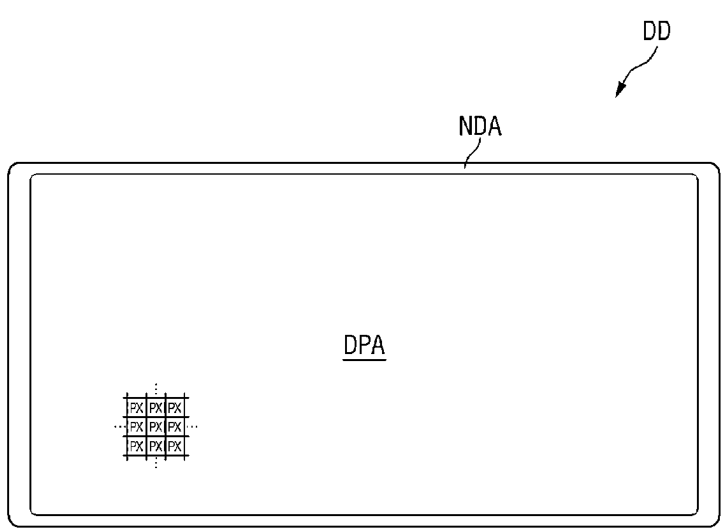
(57) **ABSTRACT**

An apparatus for manufacturing a display device includes a chamber, a laser module including a plurality of laser generators which respectively generates a plurality of raw laser beams using a solid material as a medium, a beam coupling module which combines the plurality of raw laser beams into a plurality of coupling beams, a beam mixing unit which converts the plurality of coupling beams into a plurality of mixing beams, a micro smoother which induces vibration movements of the plurality of mixing beams, an optical module which converts the plurality of mixing beams passed through the micro smoother into a final laser beam, and directs the final laser beam to the chamber, and a seal box disposed between the optical module and the chamber.

20 Claims, 26 Drawing Sheets



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* cited by examiner

FIG. 1

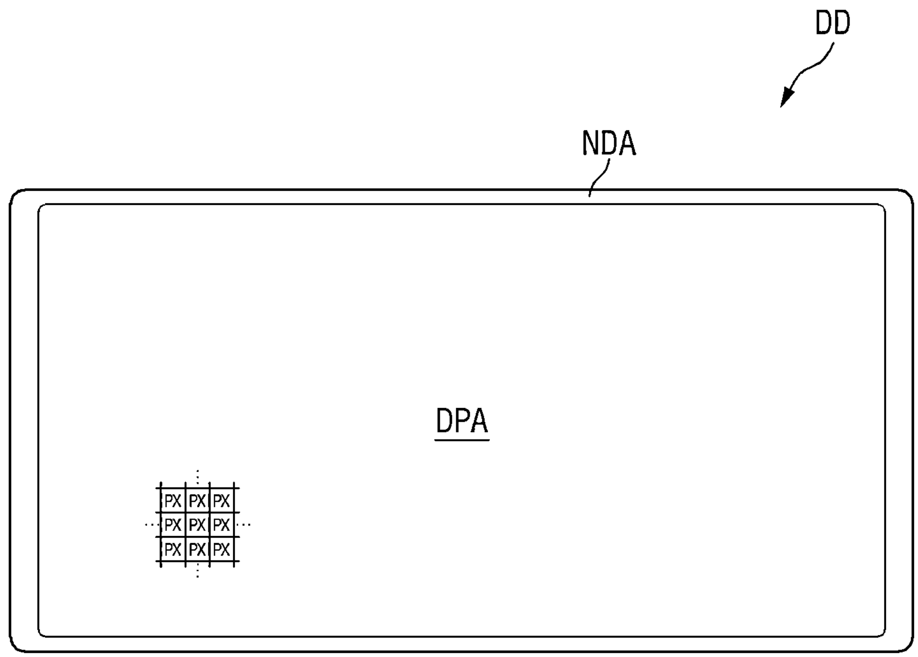
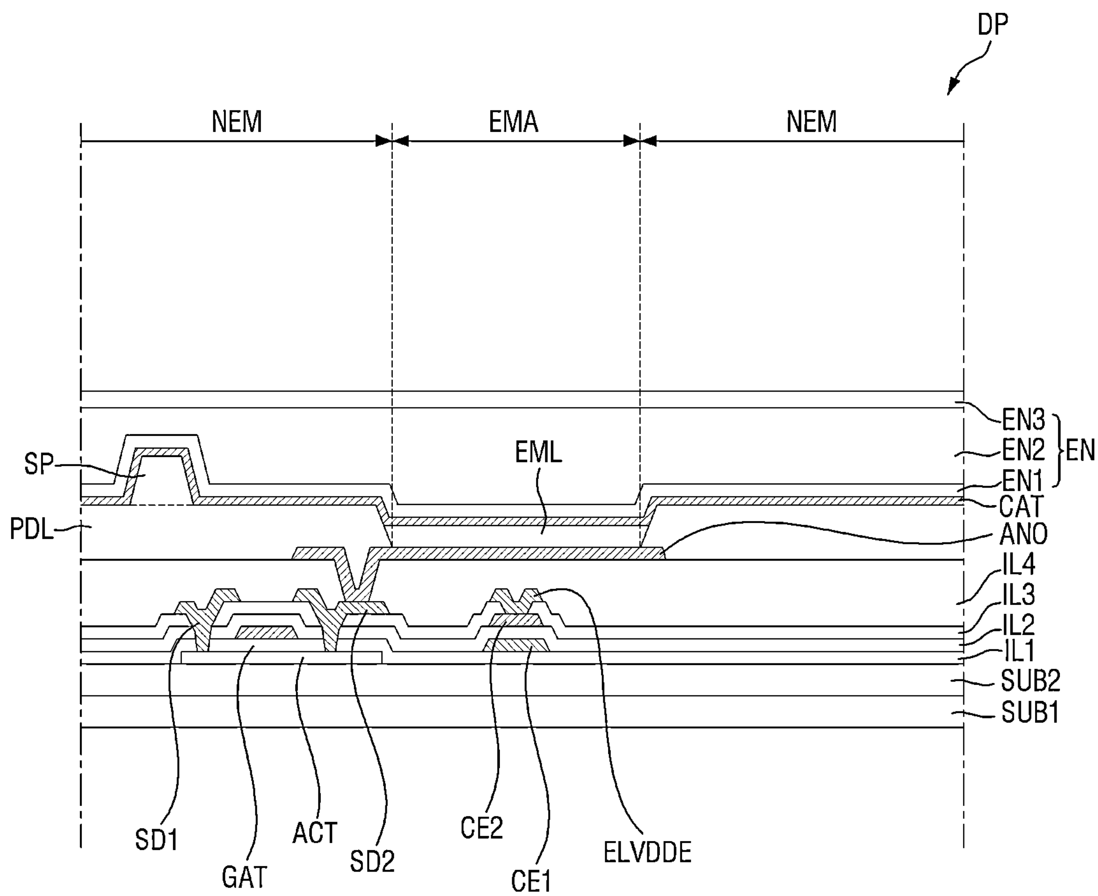


FIG. 2



GCL : GAT, CE1
GCL2 : CE2
DCL : SD1, SD2, ELVDDE

FIG. 3A

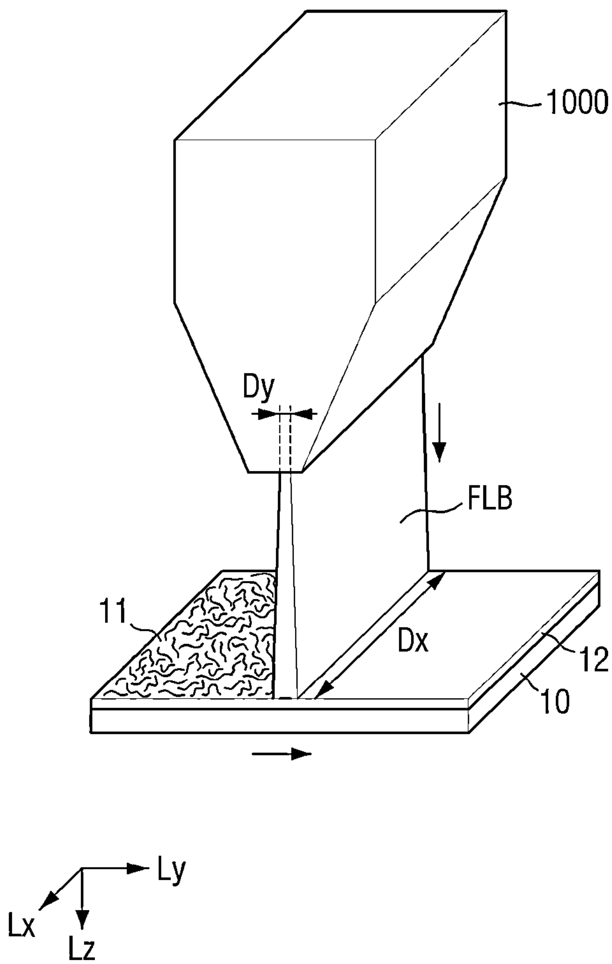


FIG. 3B

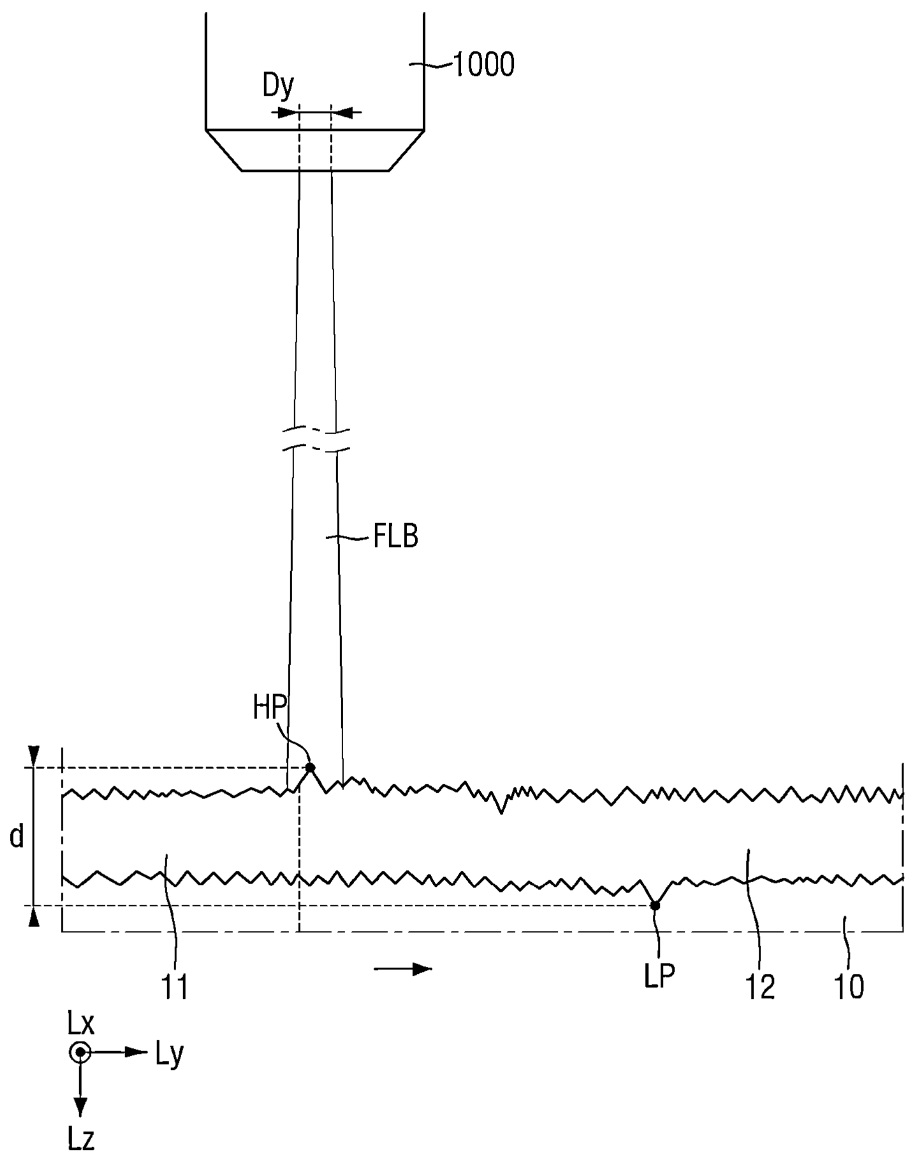


FIG. 4A

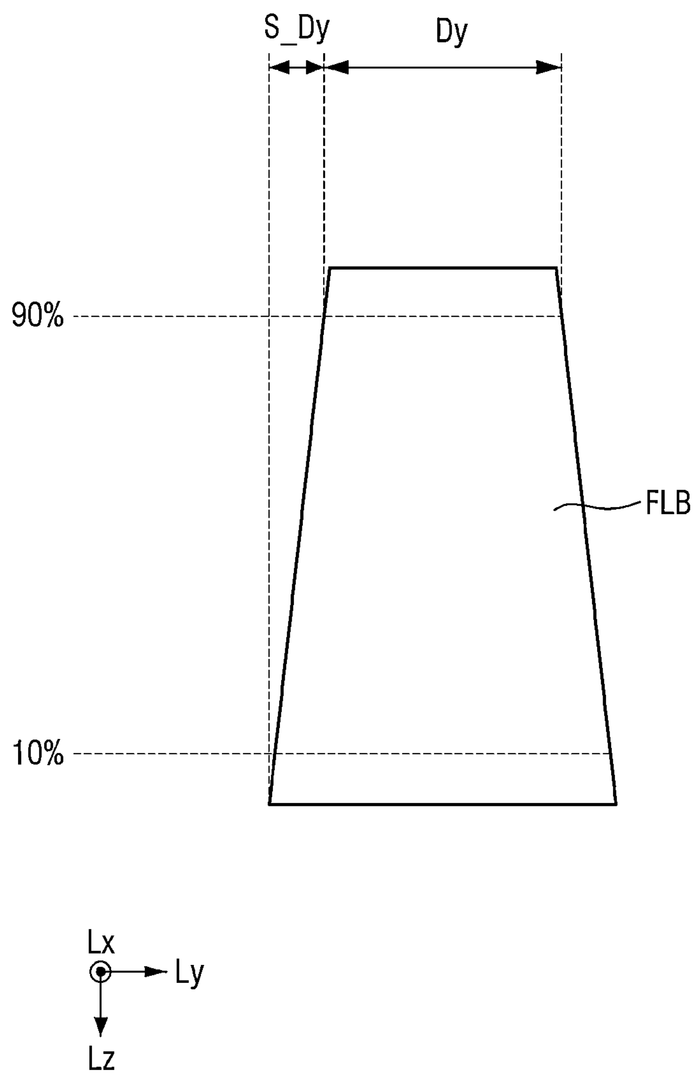


FIG. 4B

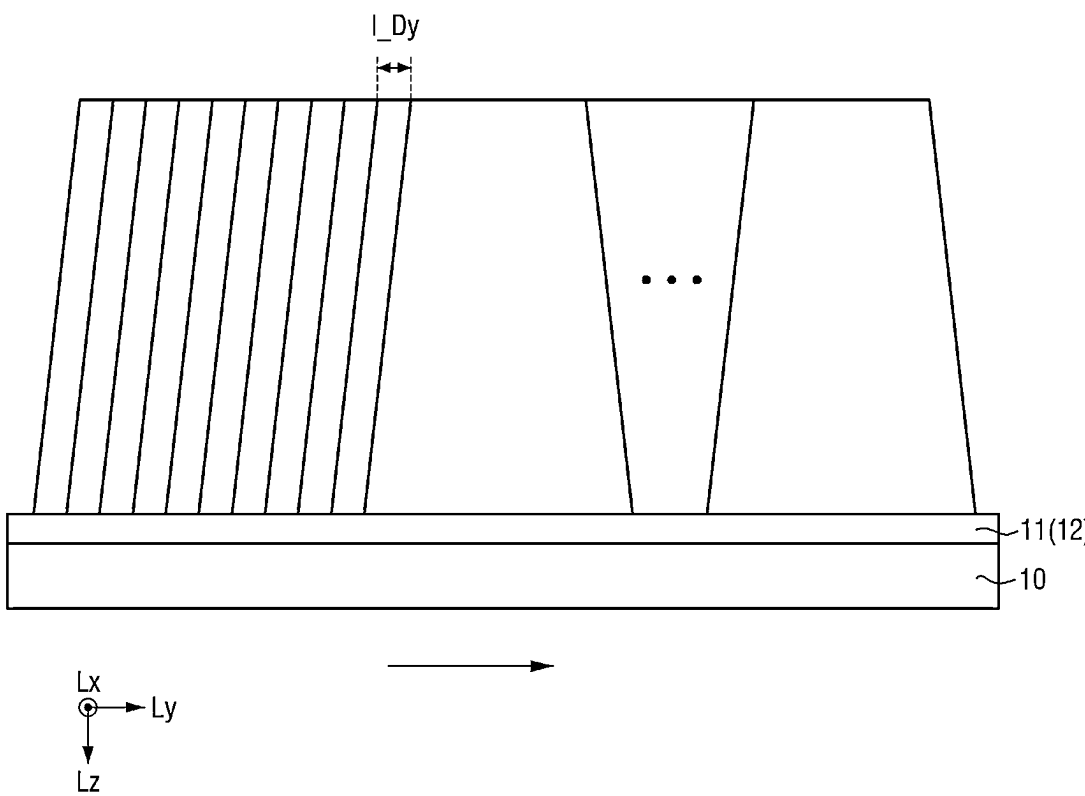


FIG. 5

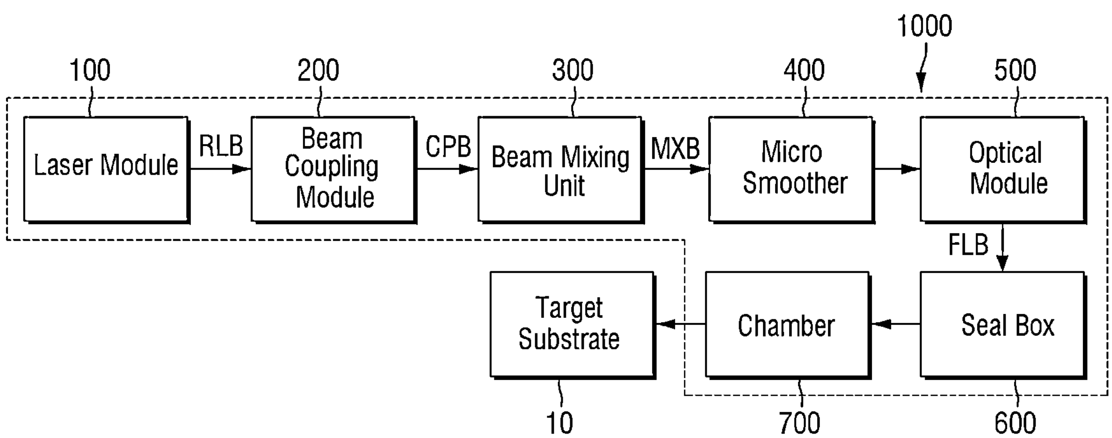

1000
100
200
300
400
500
Laser Module
RLB
Beam Coupling Module
CPB
Beam Mixing Unit
MXB
Micro Smoother
Optical Module
FLB
Target Substrate
Chamber
Seal Box
10
700
600

FIG. 6

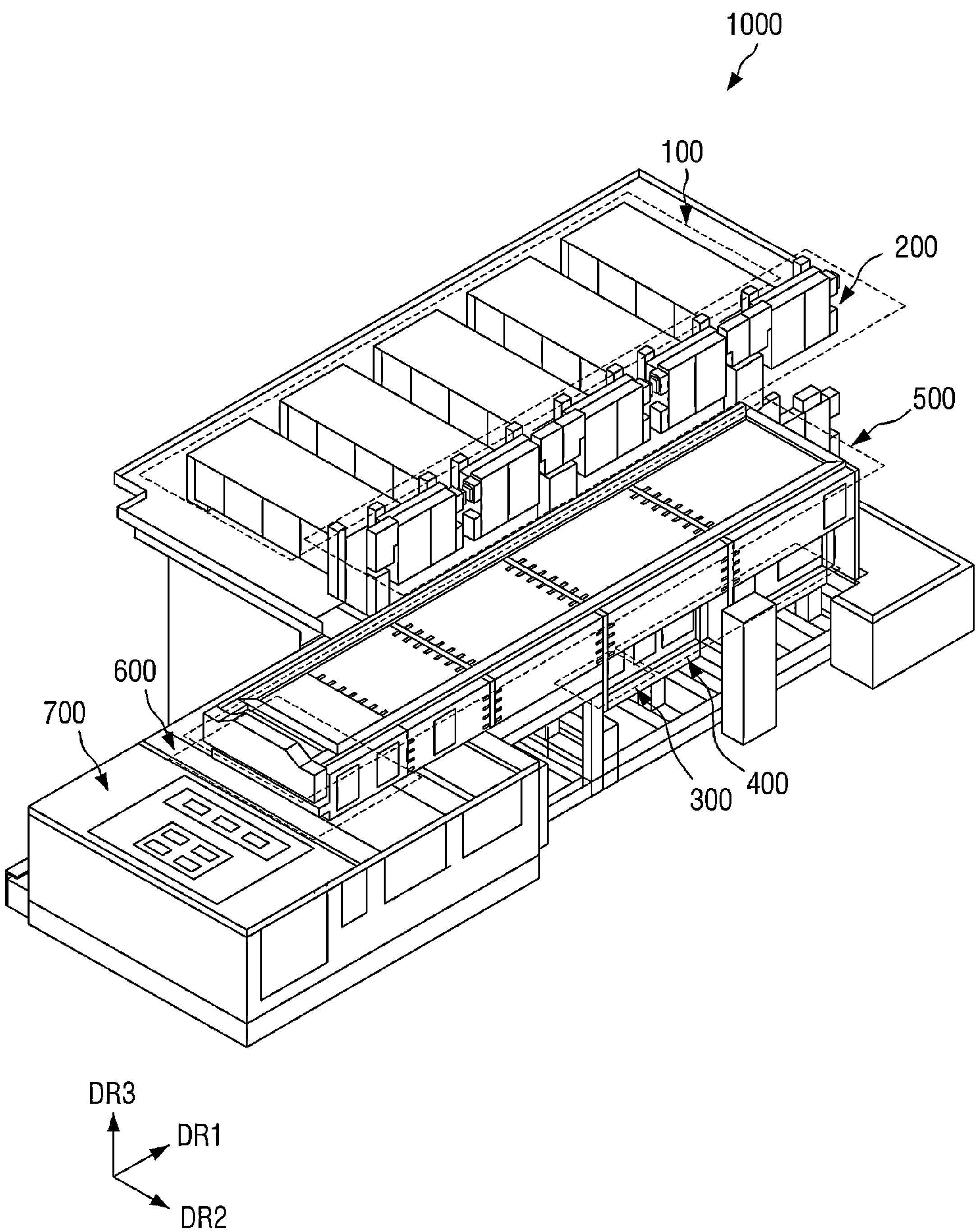
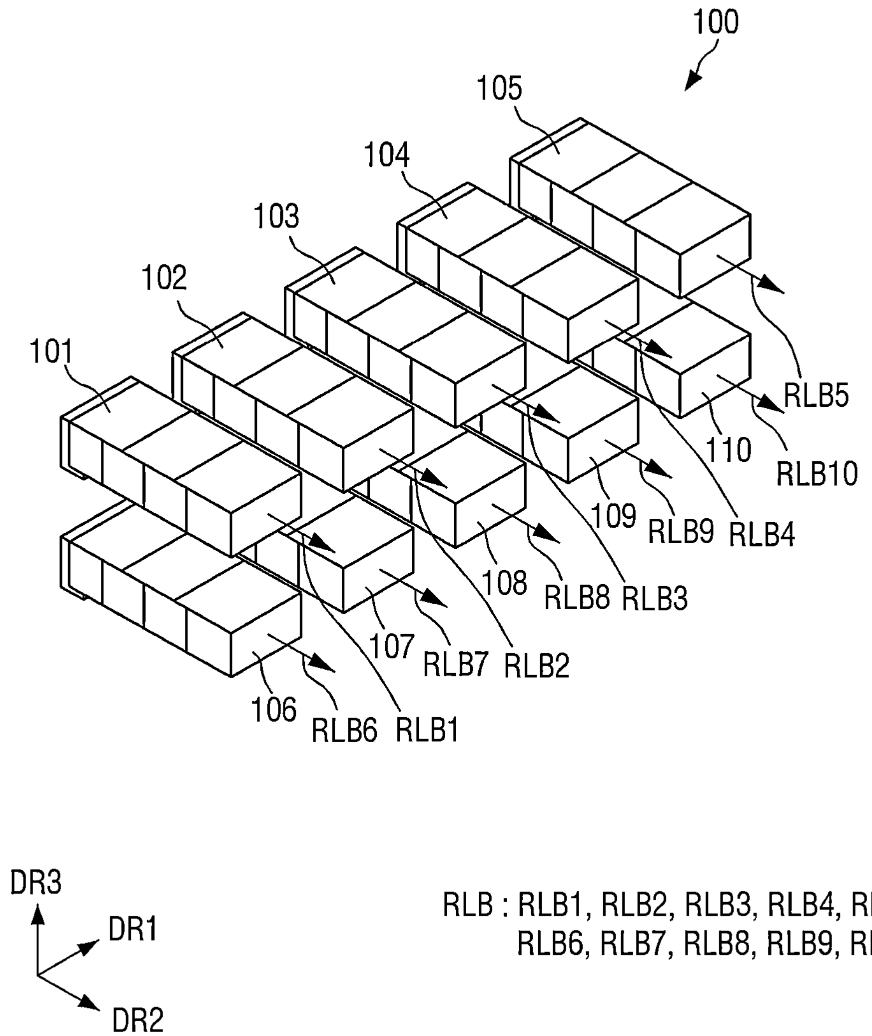


FIG. 7



RLB : RLB1, RLB2, RLB3, RLB4, RLB5,
RLB6, RLB7, RLB8, RLB9, RLB10

FIG. 8

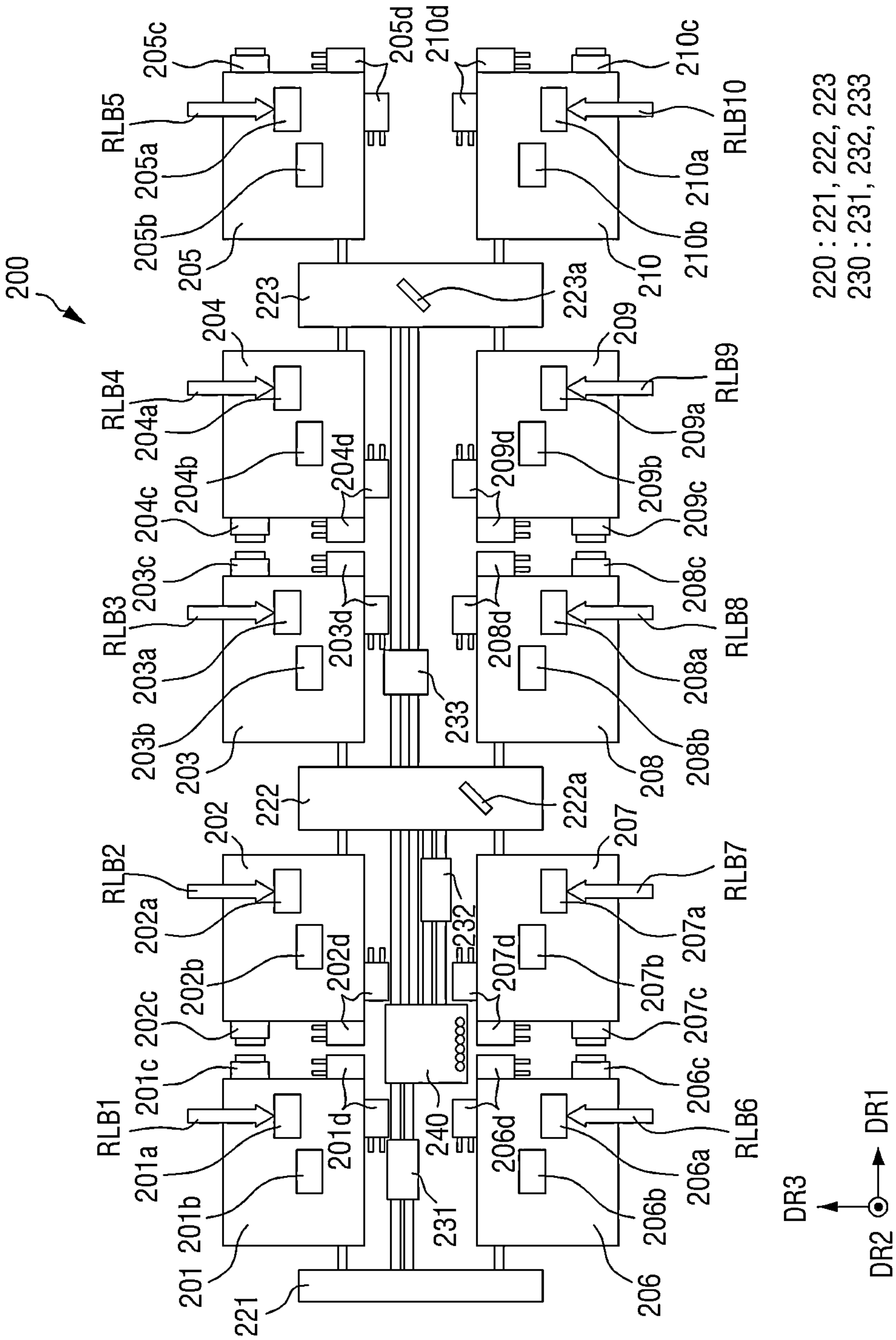
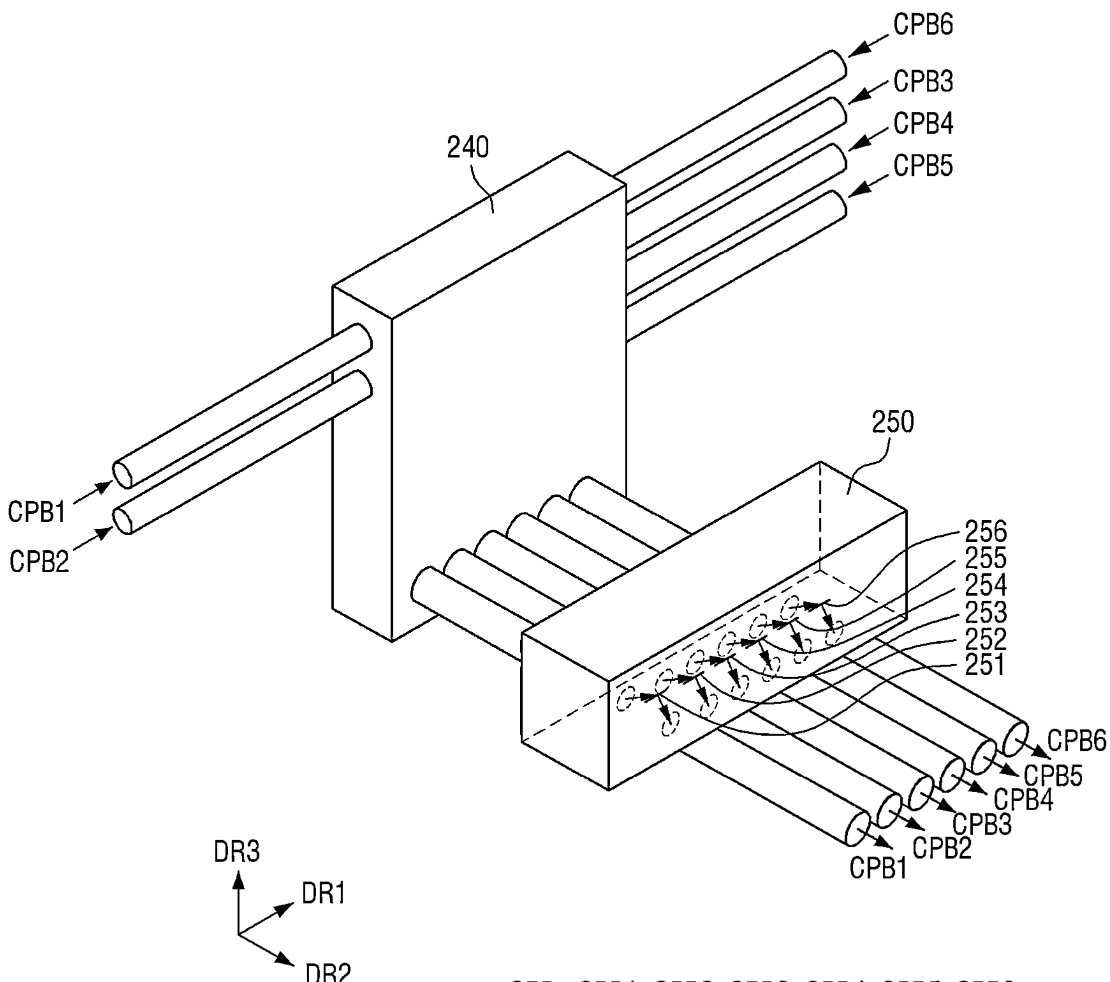

200
201 201a 201b 201c 201d RLB1
202 202a 202b 202c 202d RLB2
203 203a 203b 203c 203d RLB3
204 204a 204b 204c 204d RLB4
205 205a 205b 205c 205d RLB5
206 206a 206b 206c 206d RLB6
207 207a 207b 207c 207d RLB7
208 208a 208b 208c 208d RLB8
209 209a 209b 209c 209d RLB9
210 210a 210b 210c 210d RLB10
221 222 222a 223 223a
231 232 233 240
220 : 221, 222, 223
230 : 231, 232, 233
DR1 DR2 DR3

FIG. 9



CPB : CPB1, CPB2, CPB3, CPB4, CPB5, CPB6

FIG. 10

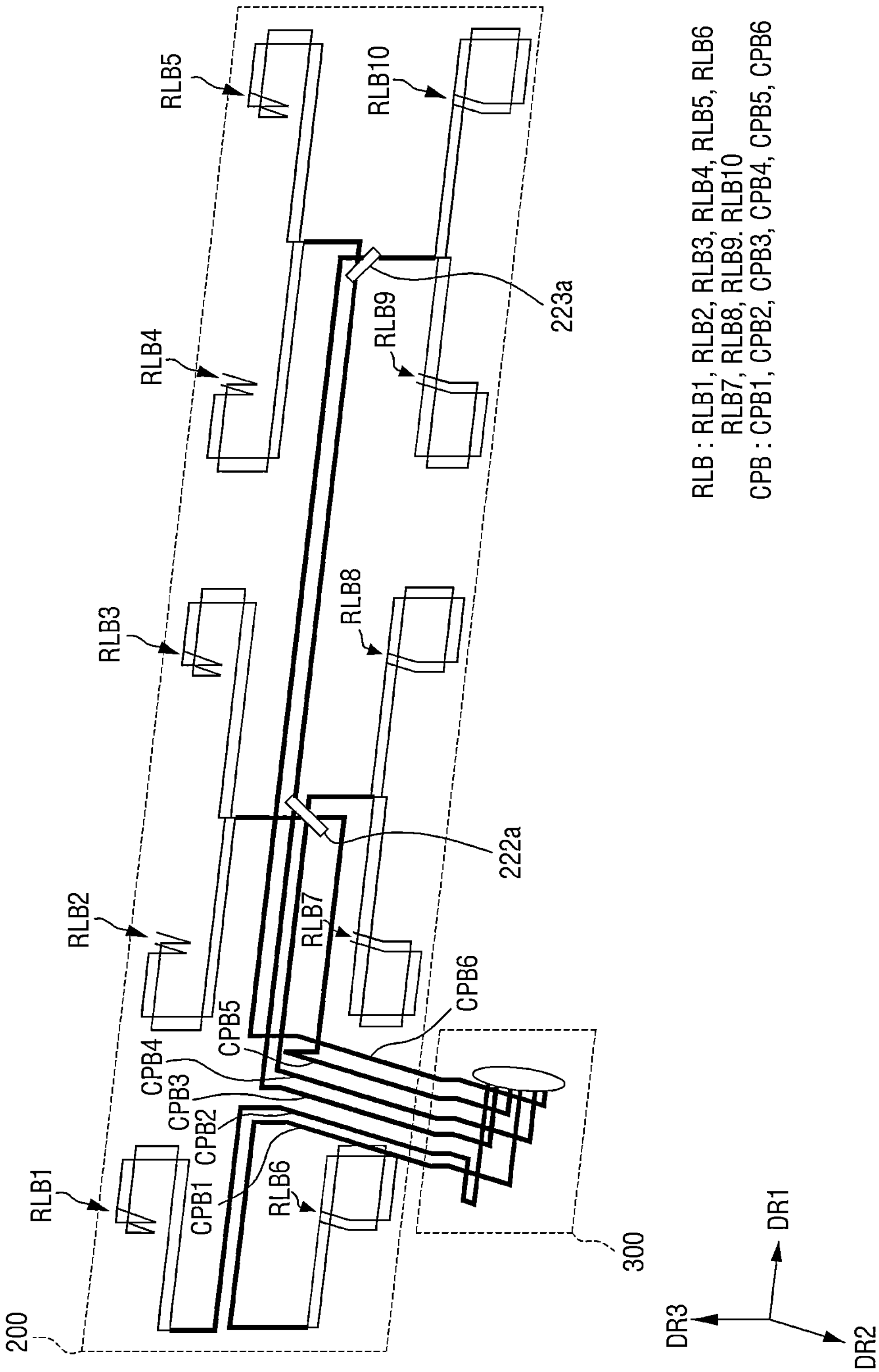
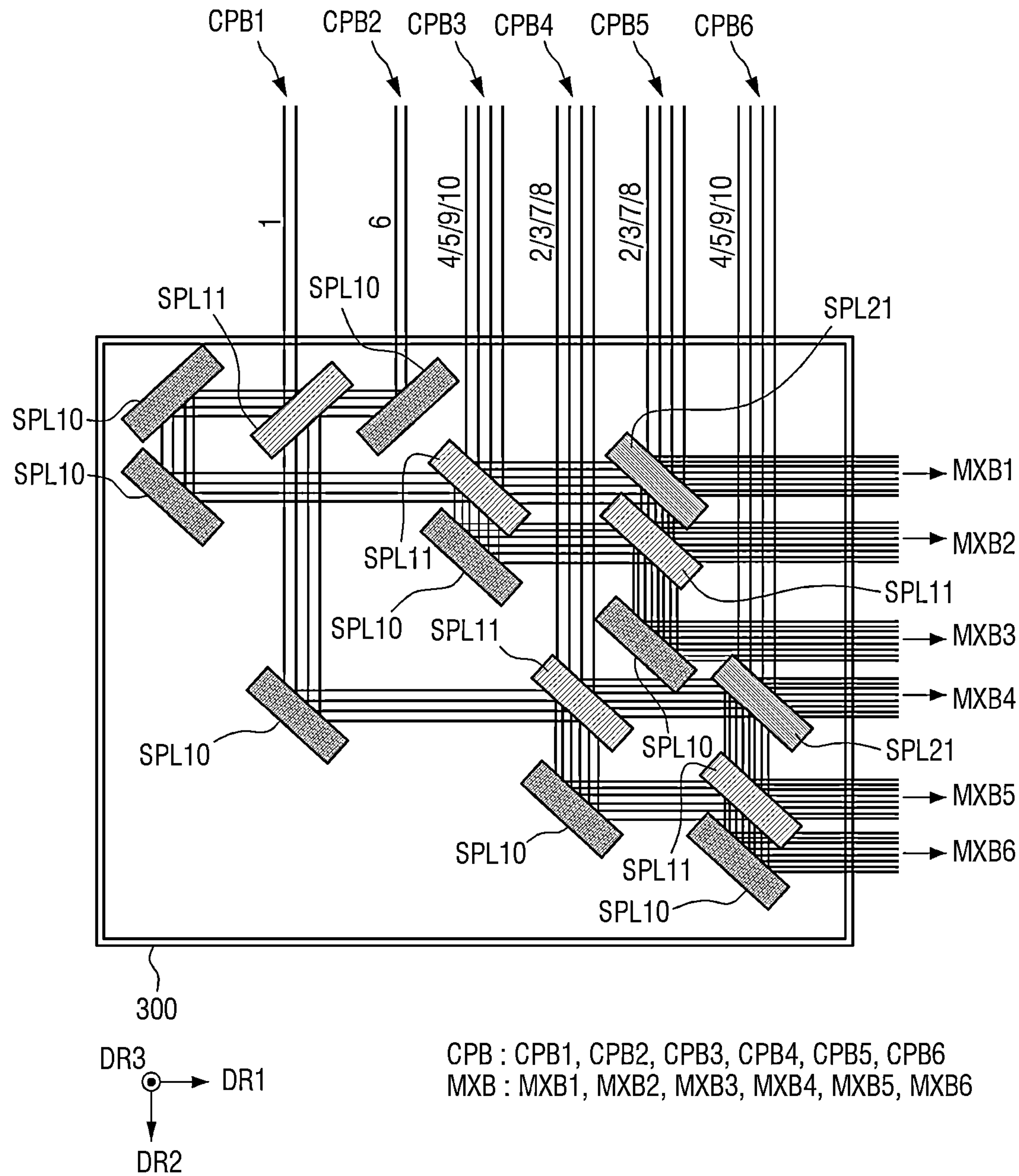
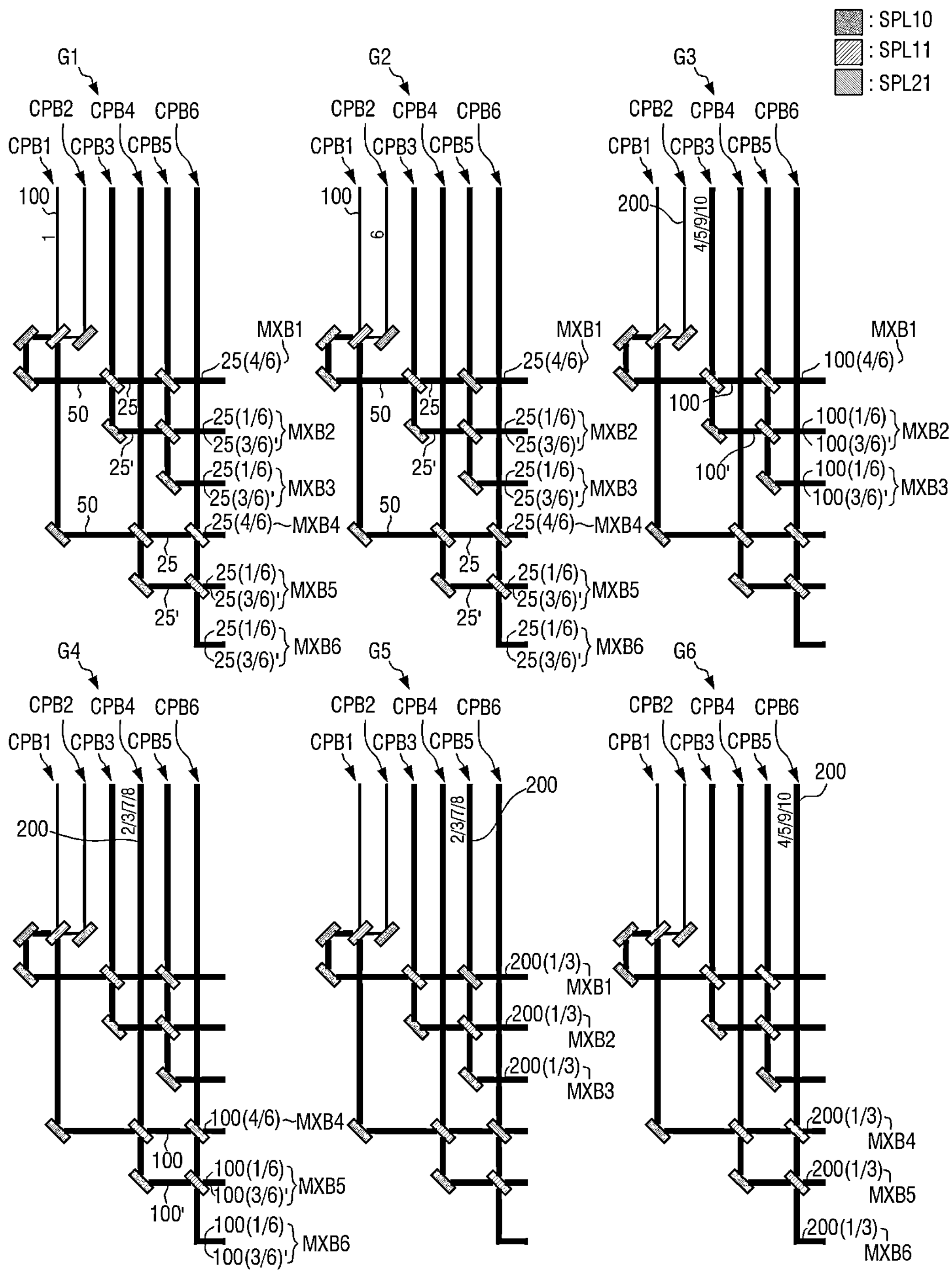

RLB5
RLB10
223a
RLB4
RLB9
RLB3
RLB8
222a
RLB2
RLB7
CPB6
CPB5
CPB4
CPB3
CPB2
CPB1
RLB1
RLB6
200
300
DR1
DR2
DR3
RLB : RLB1, RLB2, RLB3, RLB4, RLB5, RLB6
RLB7, RLB8, RLB9. RLB10
CPB : CPB1, CPB2, CPB3, CPB4, CPB5, CPB6

FIG. 11



CPB : CPB1, CPB2, CPB3, CPB4, CPB5, CPB6
MXB : MXB1, MXB2, MXB3, MXB4, MXB5, MXB6

FIG. 12



: SPL10
: SPL11
: SPL21
G1
G2
G3
G4
G5
G6
CPB1
CPB2
CPB3
CPB4
CPB5
CPB6
100
200
1
6
4/5/9/10
2/3/7/8
MXB1
MXB2
MXB3
MXB4
MXB5
MXB6
25(4/6)
25(1/6)
25(3/6)'
50
25
25'
100(4/6)
100(1/6)
100(3/6)'
100
100'
200(1/3)

FIG. 13

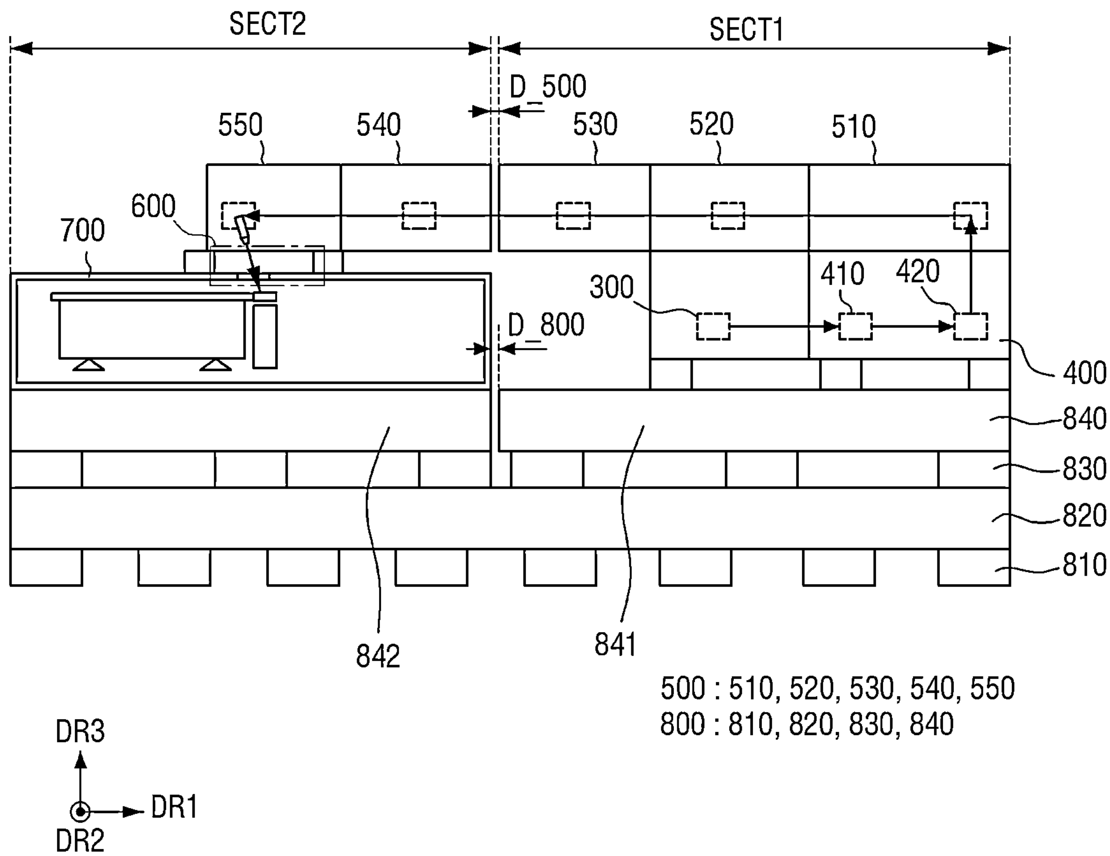

SECT2
SECT1
D_500
550
540
530
520
510
600
700
300
410
420
D_800
400
840
830
820
810
842
841
500 : 510, 520, 530, 540, 550
800 : 810, 820, 830, 840
DR3
DR1
DR2

FIG. 14

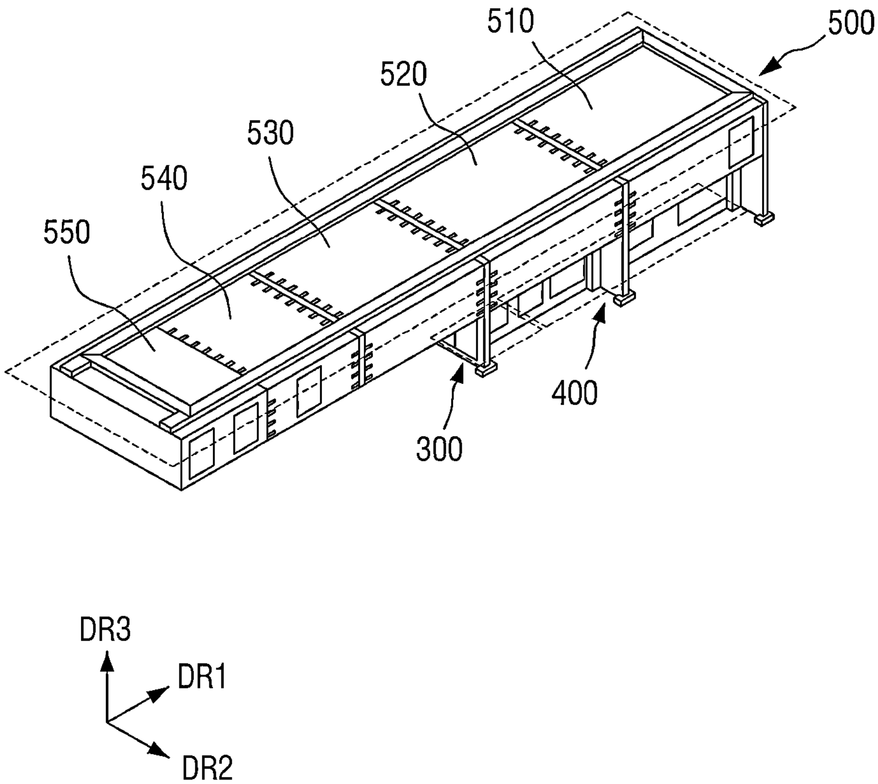


FIG. 15

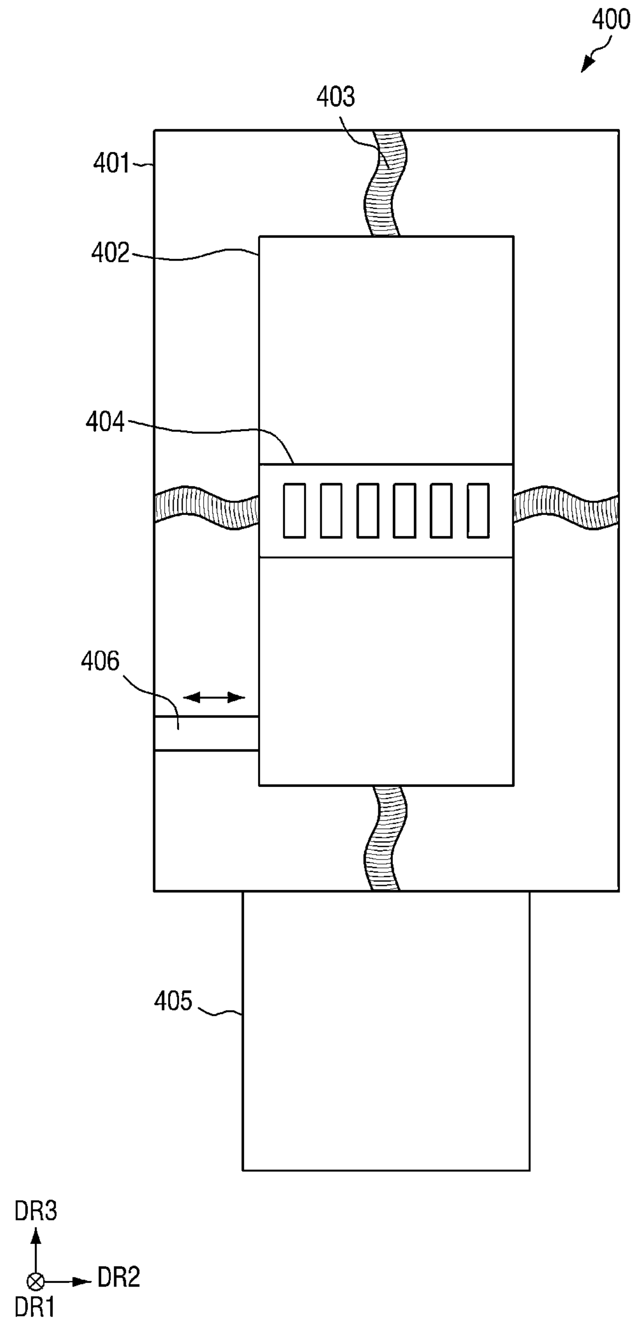


FIG. 16

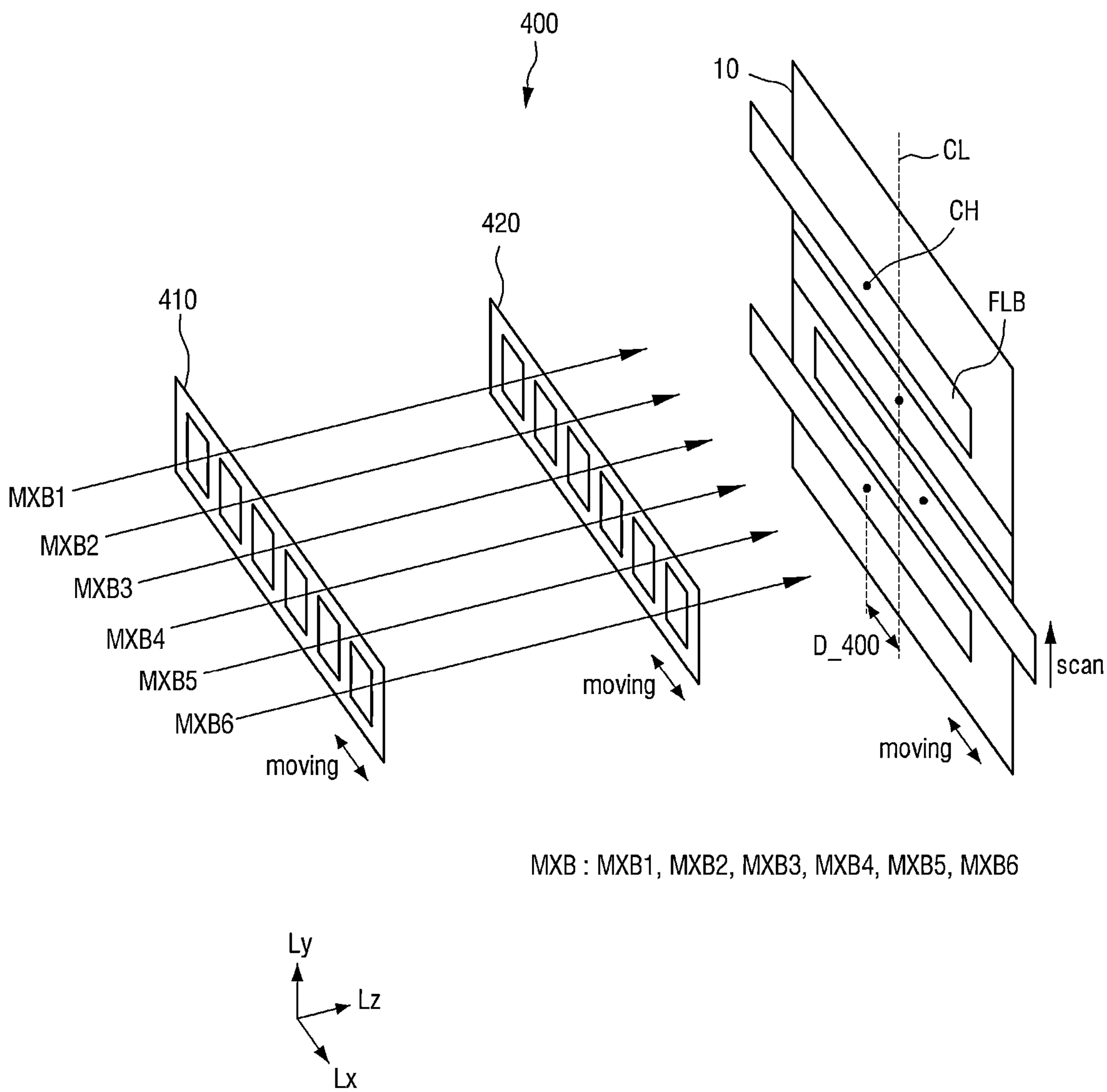

400
10
CL
CH
420
FLB
410
MXB1
MXB2
MXB3
MXB4
MXB5
MXB6
moving
moving
D_400
scan
moving
MXB : MXB1, MXB2, MXB3, MXB4, MXB5, MXB6
Ly
Lz
Lx

FIG. 17

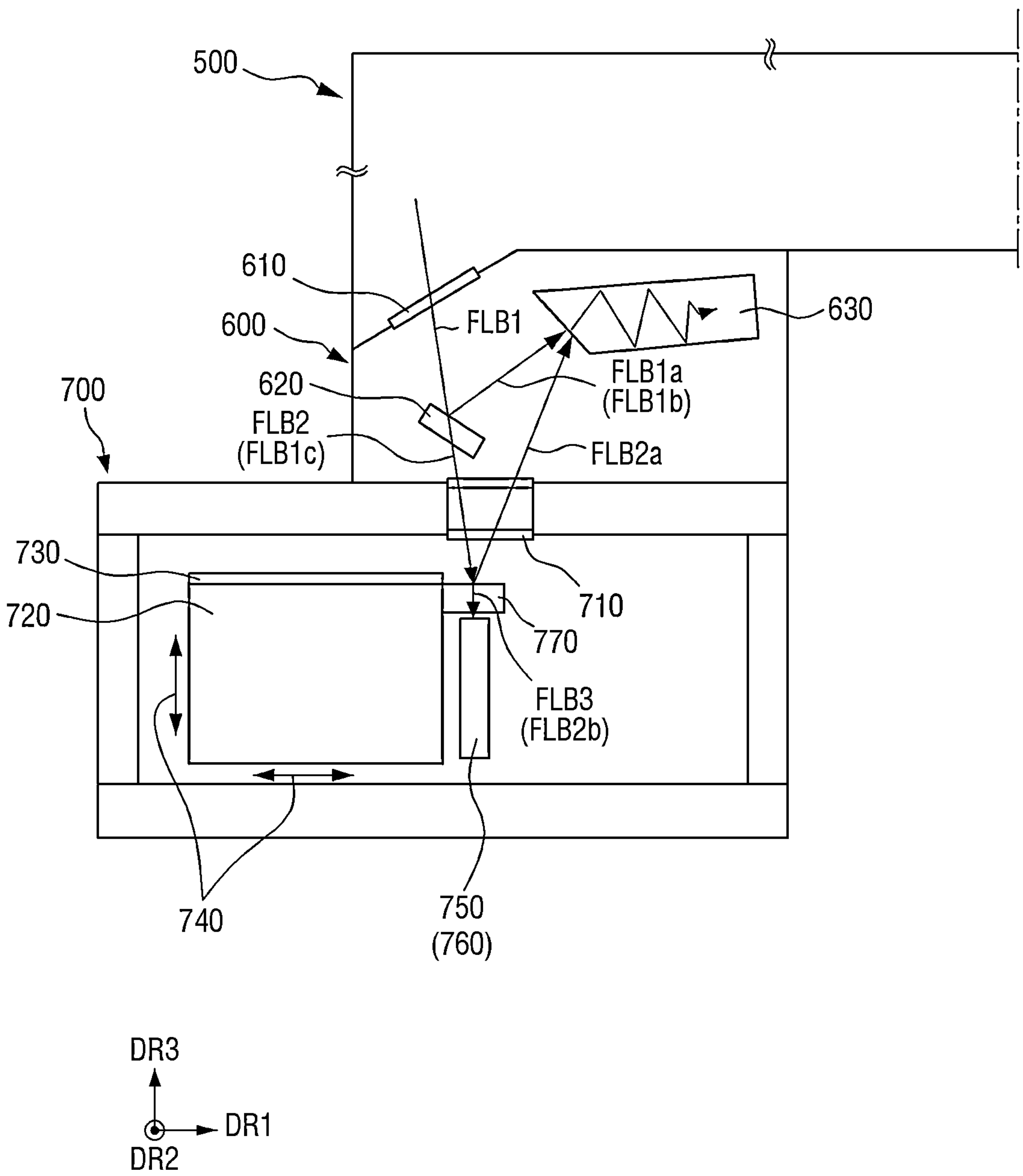

500
610
600
620
FLB2
(FLB1c)
700
FLB1
630
FLB1a
(FLB1b)
FLB2a
730
720
710
770
FLB3
(FLB2b)
740
750
(760)
DR3
DR1
DR2

FIG. 18

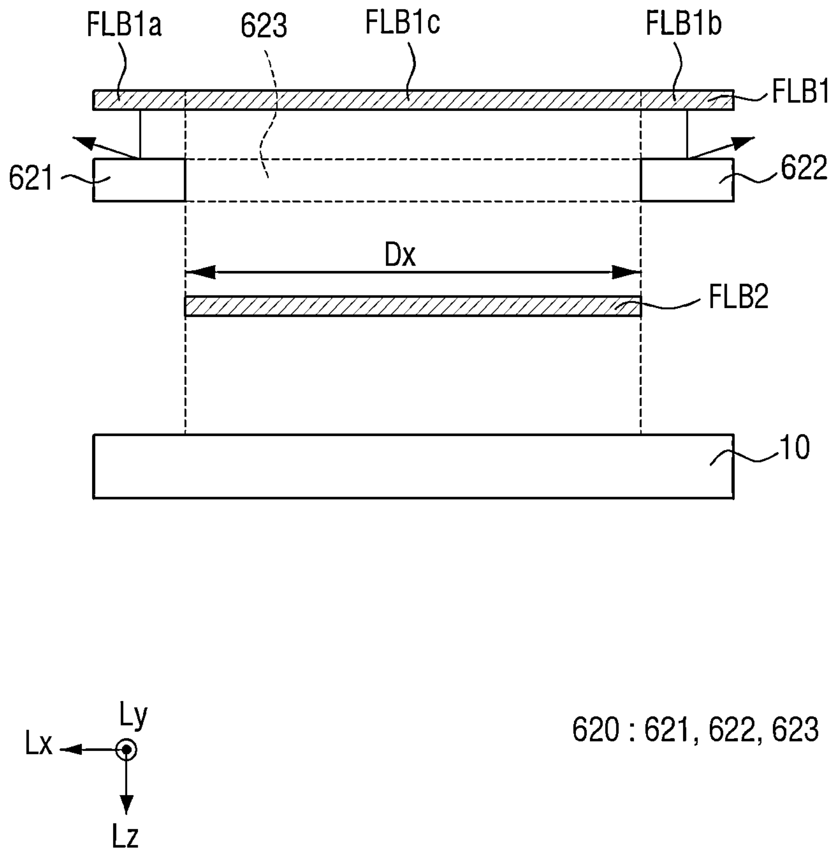

FLB1a
623
FLB1c
FLB1b
FLB1
621
622
Dx
FLB2
10
Ly
Lx
Lz
620 : 621, 622, 623

FIG. 19

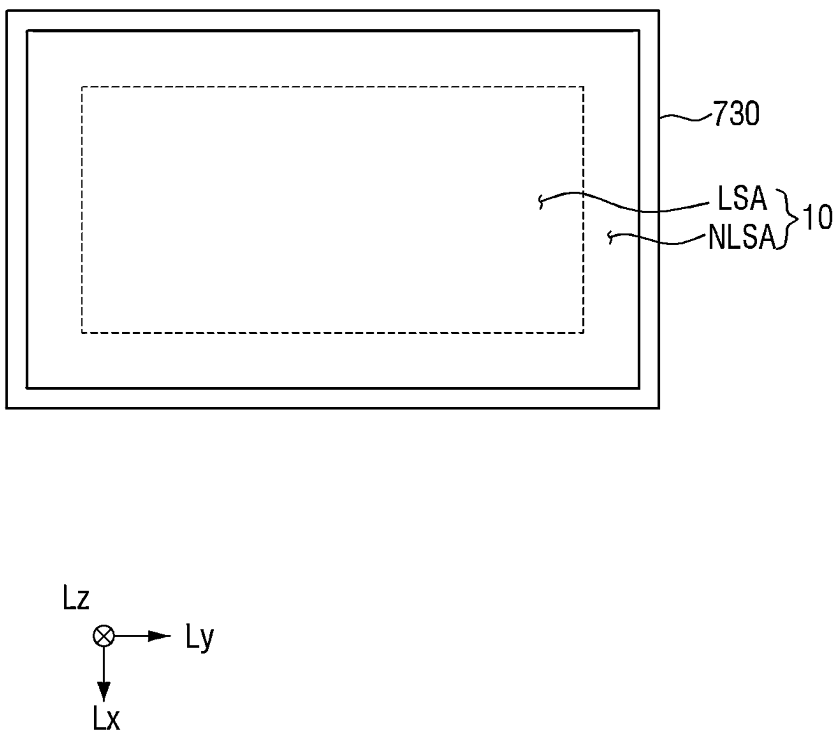


FIG. 20

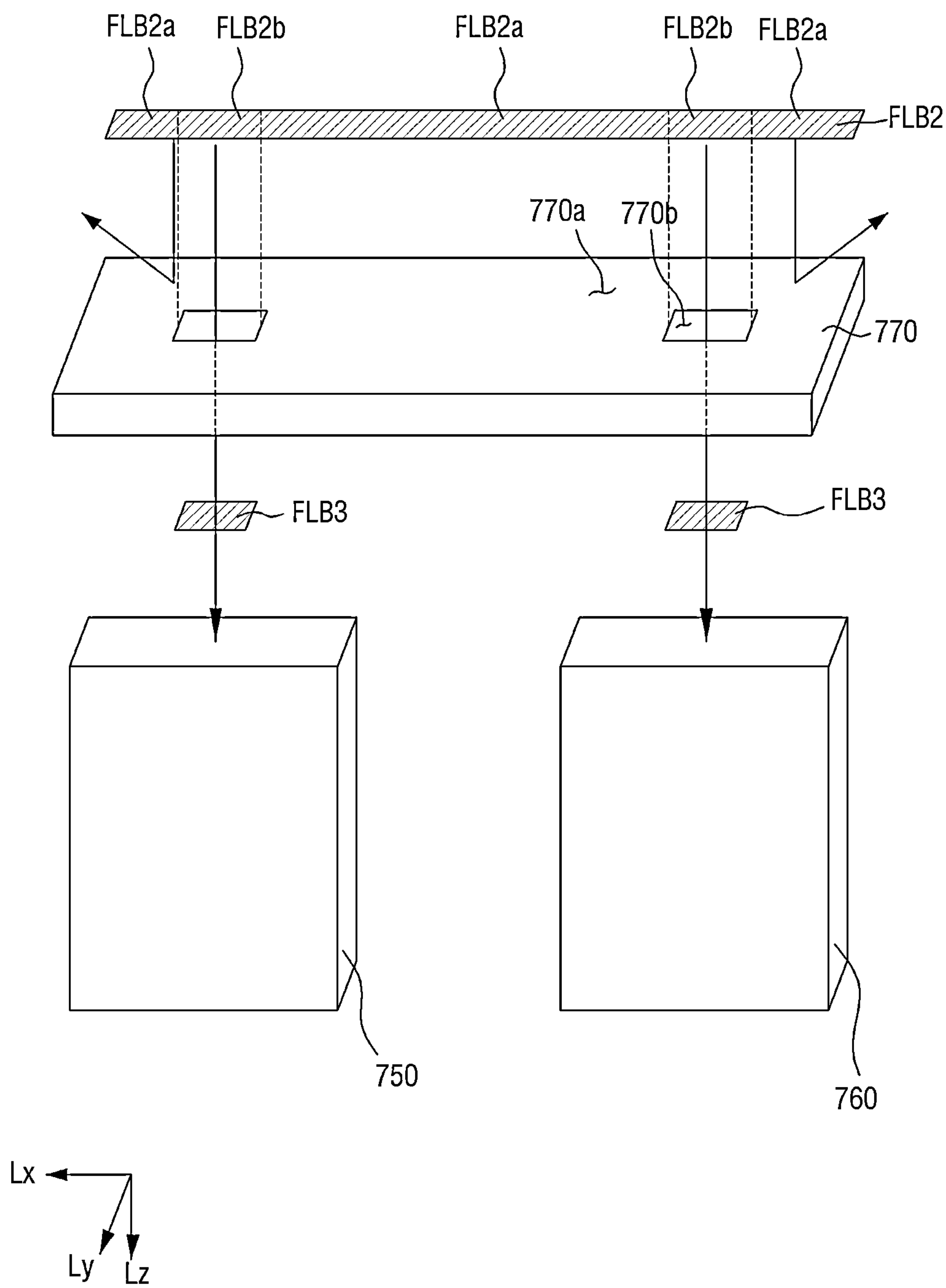

FLB2a
FLB2b
FLB2a
FLB2b
FLB2a
FLB2
770a
770b
770
FLB3
FLB3
750
760
Lx
Ly
Lz

FIG. 21

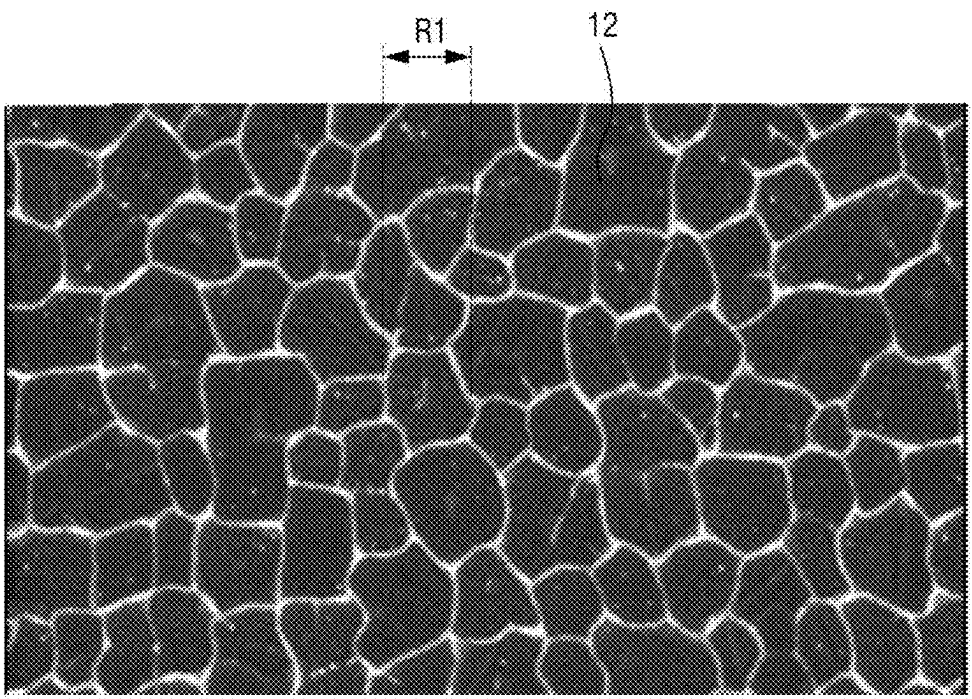


FIG. 22

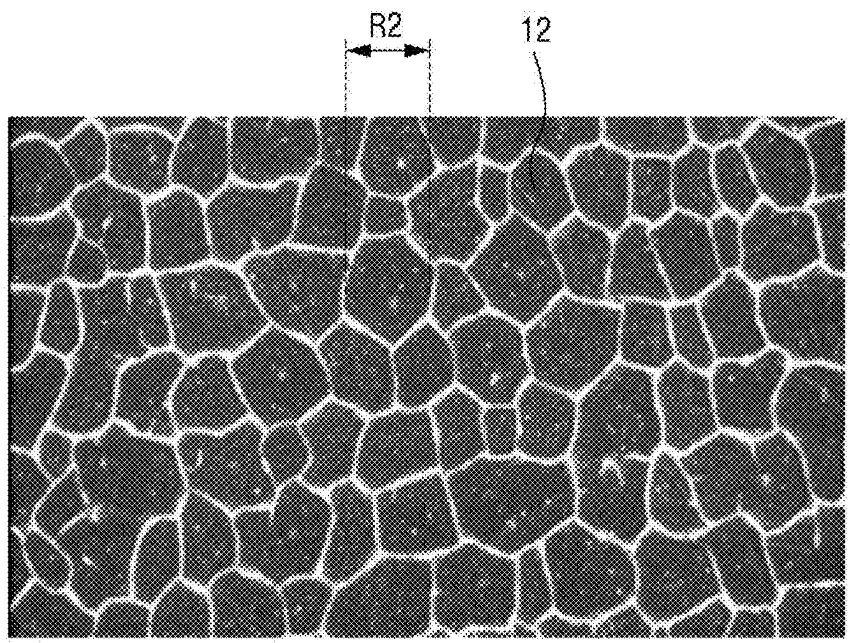


FIG. 23

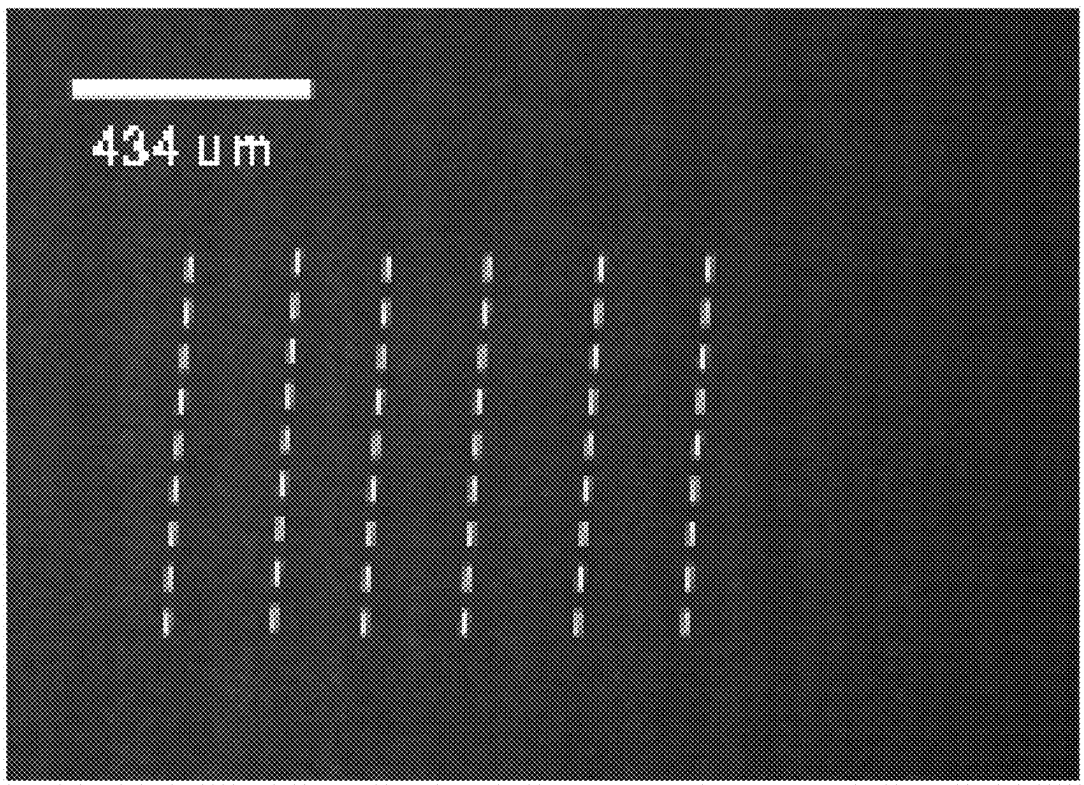
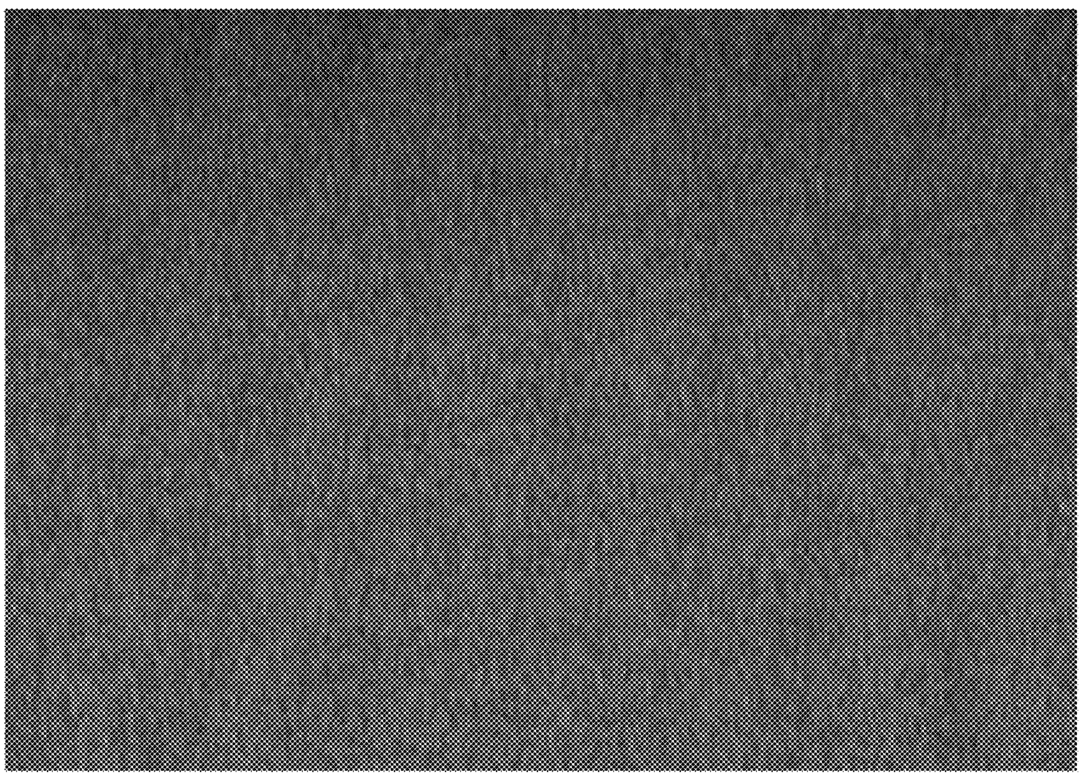


FIG. 24



APPARATUS FOR MANUFACTURING DISPLAY DEVICE

APPARATUS FOR MANUFACTURING DISPLAY DEVICE

This application claims priority to Korean Patent Application No. 10-2023-0170769, filed on Nov. 30, 2023, and all the benefits accruing therefrom under 35 U.S.C. § 119, the content of which in its entirety is herein incorporated by reference.

BACKGROUND

1. Field

Embodiments of the disclosure relate to an apparatus for manufacturing a display device.

2. Description of the Related Art

With the advance of information-oriented society, demands on display devices for displaying images are increasing in various fields. The display device may be a display device such as a liquid crystal display, a field emission display or a light emitting display. The light emitting display may include an organic light emitting display device including an organic light emitting diode as a light emitting element or an inorganic light emitting display device including an inorganic light emitting diode as a light emitting element.

The display device may control the emission status and intensity of each pixel using a thin film transistor. The thin film transistor includes a semiconductor layer, a gate electrode, source/drain electrodes, and the like. Polycrystalline silicon (poly-Si), which is a crystallization of amorphous silicon (a-Si), is commonly used in the semiconductor layer. One method of annealing to crystallize amorphous silicon (a-Si) into polysilicon (p-Si) is laser annealing, which involves irradiating the amorphous silicon (a-Si) with a laser beam to crystallize it.

SUMMARY

Embodiments of the disclosure provide an apparatus for manufacturing a display device with improved efficiency and reliability of an annealing process.

However, embodiments of the disclosure are not restricted to the one set forth herein. The above and other features of embodiments of the disclosure will become more apparent to one of ordinary skill in the art to which the disclosure pertains by referencing the detailed description of the disclosure given below.

According to an embodiment of the disclosure, an apparatus for manufacturing a display device includes a chamber, a laser module including a plurality of laser generators which respectively generates a plurality of raw laser beams using a solid material as a medium, a beam coupling module which combines the plurality of raw laser beams into a plurality of coupling beams, a beam mixing unit which convert the plurality of coupling beams into a plurality of mixing beams, a micro smoother which induces vibration movements of the plurality of mixing beams, an optical module which converts the plurality of mixing beams that have passed through the micro smoother into a final laser beam, and directs the final laser beam to the chamber, and a seal box disposed between the optical module and the chamber.

In an embodiment, a major axis size of the final laser beam may be in a range of about 1,000 millimeters (mm) to about 2,000 mm, a minor axis size of the final laser beam may be in a range of about 95 micrometers (μm) to about 110 μm, and a horizontal length of a lateral inclined surface of the final laser beam may be in a range of about 30 μm to about 45 μm.

In an embodiment, a total laser output of the laser module may be in a range of about 5,760 watts (W) to about 9,000 W, an output of each of the plurality of laser generators may be in a range of about 600 watts W to about 900 W, and an oscillation frequency of each of the plurality of laser generators may be in a range of about 5,000 hertz (Hz) to about 15,000 Hz.

In an embodiment, each of the plurality of raw laser beams may be a laser beam in an ultraviolet region, and a wavelength of the plurality of raw laser beams may be in a range of about 337 nanometers (nm) to about 357 nm.

In an embodiment, the beam coupling module may include a shutter which receives the plurality of raw laser beams, and a photodiode and a beam monitor which monitors the plurality of raw laser beams, where an operation speed of the shutter may be in a range of about 20 milliseconds (ms) to about 100 ms.

In an embodiment, the beam coupling module may include a scraper unit which generates a plurality of coupling beams, and groups the plurality of raw laser beams to be included in the plurality of coupling beams, a group telescope unit which adjusts a degree of diffusion of the plurality of coupling beams, and a group coupling unit which provides the plurality of coupling beams to the beam mixing unit.

In an embodiment, the scraper unit may include a pre-mixer, and the pre-mixer may transmit and reflect the plurality of coupling beams which are incident at a ratio of 1:1.

In an embodiment, the beam coupling module may further include a pulse extender module disposed between the group coupling unit and the beam mixing unit, and to the pulse extender module may extend a pulse duration of the final laser beam.

In an embodiment, the beam mixing unit may include a reflection mirror which reflects the plurality of incident coupling beams incident thereto, a first splitter which transmits and reflects the plurality of incident coupling beams at a ratio of 1:1, and a second splitter which transmits and reflects the plurality of incident coupling beams at a ratio of 2:1.

In an embodiment, the beam mixing unit may generate the plurality of mixing beams by combining the plurality of coupling beams at a same ratio.

In an embodiment, the micro smoother may provide a driving force to regularly or irregularly vibrate a hit point of the final laser beam with respect to a direction perpendicular to a scan direction of the final laser beam.

In an embodiment, the micro smoother may include at least one selected from, a first micro smoother comprising a telescope lens, and a second micro smoother comprising a homogenizer.

In an embodiment, the micro smoother may include a cover, a floating box disposed in an inner space of the cover in a floating state, and an elastic tube connecting the floating box to the cover.

In an embodiment, the apparatus may further include a first section where the micro smoother is located, a second section where the chamber is located, and a basic frame disposed below the chamber and the micro smoother, wherein the basic frame comprises a first pedestal disposed in the first section and a second pedestal disposed in the second section, and the first pedestal and the second pedestal are disposed to be spaced apart from each other.

In an embodiment, the optical module may include a first sub-optical module disposed in the first section, and a second sub-optical module disposed in the second section, and the first sub-optical module and the second sub-optical module are disposed to be spaced apart from each other.

In an embodiment, the seal box may include a beam cutter which reflects a portion of the final laser beam and transmits a remaining portion of the final laser beam, and a beam dump which attenuates or extinguishes energy of the portion of the final laser beam reflected from the beam cutter.

In an embodiment, the beam cutter may include a reflection area which reflects the final laser beam, and a surface of the reflection area contains glass.

In an embodiment, the apparatus may further include a reflection mirror disposed on an inner surface of the beam dump, and a cooler disposed inside the beam dump.

In an embodiment, the chamber may include a meter which measures a profile or power of the final laser beam, and a measurement aperture disposed on the meter, where the measurement aperture reflects a portion of the final laser beam and transmit a remaining portion of the final laser beam.

In an embodiment, the measurement aperture may include a reflection area which reflects the final laser beam, and a transmission area which transmits the final laser beam, where the reflection area may include a high reflection (HR) coating film, and the transmission area may include an anti-reflection (AR) coating film or a hole.

In the apparatus for manufacturing the display device according to an embodiment of the disclosure, the efficiency and reliability of the annealing process may be improved.

However, effects according to the embodiments of the disclosure are not limited to those exemplified above and various other effects are incorporated herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features of embodiments of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings, in which:

FIG. **1** is a plan view illustrating a display device according to an embodiment;

FIG. **2** is a cross-sectional view showing a display panel according to an embodiment;

FIG. **3**A is a perspective view illustrating an annealing process of a display device according to an embodiment;

FIG. **3**B is a cross-sectional view illustrating an annealing process of a display device according to an embodiment;

FIG. **4**A is a front view showing the shape of a final laser beam according to an embodiment;

FIG. **4**B is a front view cumulatively illustrating the locations where the final laser beam is irradiated as the target substrate according to an embodiment moves;

FIG. **5** is a schematic block diagram showing an apparatus for manufacturing a display device according to an embodiment;

FIG. **6** is a perspective view showing an apparatus for manufacturing a display device according to an embodiment;

FIG. **7** is a perspective view showing a laser module according to an embodiment;

FIG. **8** is a cross-sectional view showing a beam coupling module according to an embodiment;

FIG. **9** is a perspective view illustrating a group coupling unit and a pulse extender module according to an embodiment;

FIG. **10** is a schematic diagram showing paths of laser beams passing through a beam coupling module and a beam mixing unit according to an embodiment;

FIG. **11** is a schematic diagram showing the path of laser beams passing through a beam mixing unit according to an embodiment;

FIG. **12** is a schematic diagram showing a process in which each coupling beam is distributed into a mixing beam in a beam mixing unit according to an embodiment;

FIG. **13** is a cross-sectional view illustrating a beam mixing unit, a micro smoother, an optical module, a seal box, a chamber, and a basic frame according to an embodiment;

FIG. **14** is a perspective view illustrating the beam mixing unit, the micro smoother, and the optical module according to an embodiment;

FIG. **15** is a cross-sectional view illustrating a micro smoother according to an embodiment;

FIG. **16** is a perspective view illustrating an operation method of a micro smoother according to an embodiment;

FIG. **17** is a cross-sectional view illustrating an optical module, a seal box, and a chamber according to an embodiment;

FIG. **18** is a cross-sectional view illustrating an operation method of a beam cutter according to an embodiment;

FIG. **19** is a plan view illustrating a fixing chuck and a target substrate according to an embodiment;

FIG. **20** is a perspective view illustrating a profile meter, a power meter, and a measurement aperture according to an embodiment;

FIG. **21** is a microscopic image illustrating particles of a polycrystalline silicon thin film according to a conventional embodiment;

FIG. **22** is a microscopic image illustrating particles of a polycrystalline silicon thin film according to an embodiment;

FIG. **23** is a photograph illustrating a macro inspection of a polycrystalline silicon thin film according to the conventional embodiment; and

FIG. **24** is a photograph illustrating a macro inspection of a polycrystalline silicon thin film according to an embodiment.

DETAILED DESCRIPTION

The invention will now be described more fully hereinafter with reference to the accompanying drawings, in which various embodiments of the invention are shown. This invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will filly convey the scope of the invention to those skilled in the art. Like reference numerals refer to like elements throughout.

It will also be understood that when a layer is referred to as being "on" another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present.

It will be understood that, although the terms "first," "second," "third" etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, "a first element," "component," "region," "layer" or "section" discussed below could be termed a second element, component, region, layer or section without departing from the teachings herein.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, "a", "an," "the," and "at least one" do not denote a limitation of quantity, and are intended to include both the singular and plural, unless the context clearly indicates otherwise. Thus, reference to "an" element in a claim followed by reference to "the" element is inclusive of one element and a plurality of the elements. For example, "an element" has the same meaning as "at least one element," unless the context clearly indicates otherwise. "At least one" is not to be construed as limiting "a" or "an." "Or" means "and/or." As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

Furthermore, relative terms, such as "lower" or "bottom" and "upper" or "top," may be used herein to describe one element's relationship to another element as illustrated in the Figures. It will be understood that relative terms are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures. For example, if the device in one of the figures is turned over, elements described as being on the "lower" side of other elements would then be oriented on "upper" sides of the other elements. The term "lower," can therefore, encompasses both an orientation of "lower" and "upper," depending on the particular orientation of the figure. Similarly, if the device in one of the figures is turned over, elements described as "below" or "beneath" other elements would then be oriented "above" the other elements. The terms "below" or "beneath" can, therefore, encompass both an orientation of above and below.

"About" or "approximately" as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, "about" can mean within one or more standard deviations, or within ±30%, 20%, 10% or 5% of the stated value.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the claims.

Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

FIG. **1** is a plan view illustrating a display device according to an embodiment.

Referring to FIG. **1**, an embodiment of a display device DD may display a moving image or a still image. The display device DD may include any electronic device including a display screen. Examples of the display device DD may include a television, a laptop computer, a monitor, a billboard, an Internet-of-Things device, a mobile phone, a smartphone, a tablet personal computer (PC), an electronic watch, a smart watch, a watch phone, a head-mounted display, a mobile communication terminal, an electronic notebook, an electronic book, a portable multimedia player (PMP), a navigation device, a game machine, a digital camera, a camcorder and the like, which include a display screen.

The shape of the display device DD may be variously modified. In an embodiment, for example, the display device DD may have a shape such as a rectangular shape elongated in a horizontal direction, a rectangular shape elongated in a vertical direction, a square shape, a quadrilateral shape with rounded corners (vertices), other polygonal shapes and a circular shape. The shape of a display area DPA of the display device DD may also be similar to the overall shape of the display device DD. In an embodiment, as shown in FIG. **1**, the display device DD and the display area DPA may have a rectangular shape elongated in the horizontal direction, but the disclosure is not limited thereto.

In an embodiment, the display device DD may include the display area DPA and a non-display area NDA. The display area DPA is an area where a screen can be displayed, and the non-display area NDA is an area where a screen is not displayed. The display area DPA may also be referred to as an active region, and the non-display area NDA may also be referred to as a non-active region. The display area DPA may substantially occupy the center of the display device DD.

The display area DPA may include a plurality of pixels PX. The plurality of pixels PX may be arranged in a matrix form.

The non-display area NDA may be disposed around the display area DPA. The non-display area NDA may completely or partially surround the display area DPA. The display area DPA may have a rectangular shape, and the non-display area NDA may be disposed adjacent to four sides of the display area DPA.

FIG. **2** is a cross-sectional view showing a display panel according to an embodiment.

Referring to FIG. **2**, an embodiment of the display device DD may include a display panel DP. The display panel DP displays a screen or an image and may be, for example, a self-luminous display panel, such as an organic light emitting display panel, an inorganic light emitting display panel, a quantum dot light emitting display panel, a micro light emitting diode (LED) display panel, a nano LED display panel, a plasma display panel, a field emission display panel, or a cathode ray display panel, as well as a light receiving display panel such as a liquid crystal display panel or an electrophoretic display panel. Hereinafter, for convenience of description, embodiments where the display panel DP is an organic light emitting display panel will be described as an example, but is not limited thereto.

In an embodiment, as shown in FIG. **2**, the display panel DP may include a base substrate SUB**1**, a buffer layer SUB**2**, a semiconductor layer ACT, a first insulating layer IL**1**, a first gate conductive layer GCL**1**, a second insulating layer IL**2**, a second gate conductive layer GCL**2**, a third insulating layer IL**3**, a data conductive layer DCL, a fourth insulating layer IL**4**, an anode electrode ANO, a pixel defining layer PDL including an opening exposing the anode electrode ANO, a light emitting layer EML disposed in the opening of the pixel defining layer PDL, a cathode electrode CAT disposed on the light emitting layer EML and the pixel defining layer PDL, and a thin film encapsulation layer EN disposed on the cathode electrode CAT. Each of the layers described above may consist of (or defined by) a single layer, or a stack of multiple layers. Another layer may be further disposed between the respective layers.

The base substrate SUB**1** may support respective layers disposed thereon. The base substrate SUB**1** may include or be made of an insulating material such as a polymer resin or an inorganic material such as glass or quartz.

The buffer layer SUB**2** is disposed on the base substrate SUB**1**. The buffer layer SUB**2** may include silicon nitride, silicon oxide, silicon oxynitride, or the like.

The semiconductor layer ACT may be disposed on the buffer layer SUB**2**. The semiconductor layer ACT may form a channel of a thin film transistor of a pixel. The semiconductor layer ACT may include polycrystalline silicon crystallized by a display device manufacturing apparatus **1000** (see FIG. **3**A), which will be described later. The semiconductor layer ACT may be formed by crystallizing an entire amorphous silicon thin film **11** (see FIG. **3**A) disposed on a target substrate **10** (see FIG. **3**A) and then patterning the crystallized amorphous silicon thin film **11**, or may be formed by first patterning the amorphous silicon thin film **11** (see FIG. **3**A) and then crystallizing the patterned amorphous silicon thin film **11** (see FIG. **3**A). However, the disclosure is not limited thereto, and only a part of the amorphous silicon thin film **11** (see FIG. **3**A) may be crystallized, so that the semiconductor layer ACT may include both an amorphous silicon region where amorphous silicon is disposed and a polycrystalline silicon region where polycrystalline silicon is disposed.

The first insulating layer IL**1** may be disposed on the semiconductor layer ACT. The first insulating layer IL**1** may be a gate insulating layer having a gate insulating function.

The first gate conductive layer GCL**1** may be disposed on the first insulating layer IL**1**. The first gate conductive layer GCL**1** may include a gate electrode GAT of a thin transistor of a pixel, a scan line connected thereto and a first electrode CE**1** of a storage capacitor.

The second insulating layer IL**2** may be disposed on the first gate conductive layer GCL**1**. The second insulating layer IL**2** may be an interlayer insulating layer or a second gate insulating layer.

The second gate conductive layer GCL**2** may be disposed on the second insulating layer IL**2**. The second gate conductive layer GCL**2** may include a second electrode CE**2** of the storage capacitor.

The third insulating layer IL**3** may be disposed on the second gate conductive layer GCL**2**. The third insulating layer IL**3** may be an interlayer insulating layer.

The data conductive layer DCL may be disposed on the third insulating layer IL**3**. The data conductive layer DCL may include a first electrode SD**1**, a second electrode SD**2** and a first power line ELVDDE of the thin film transistor of the pixel. The first electrode SD**1** and the second electrode SD**2** of the thin film transistor may be electrically connected to a source region and a drain region of the semiconductor layer ACT via contact holes defined or formed through the third insulating layer IL**3**, the second insulating layer IL**2** and the first insulating layer IL**1**.

The fourth insulating layer IL**4** may be disposed on the data conductive layer DCL. The fourth insulating layer IL**4** may cover the data conductive layer DCL. The fourth insulating layer IL**4** may be a via layer.

The anode electrode ANO may be disposed on the fourth insulating layer IL**4**. The anode electrode ANO may be a pixel electrode provided for each pixel. The anode electrode ANO may be connected to the second electrode SD**2** of the thin film transistor via the contact hole passing through the fourth insulating layer IL**4**.

The pixel defining layer PDL may be disposed on the anode electrode ANO. The pixel defining layer PDL may be disposed on the anode electrode ANO and may define an opening exposing the anode electrode ANO. The emission area EMA and the non-emission area NEM may be distinguished by the pixel defining layer PDL and the opening thereof.

A spacer SP may be disposed on the pixel defining layer PDL. The spacer SP may serve to maintain a gap with a structure disposed thereabove.

The light emitting layer EML may be disposed on the anode electrode ANO exposed by the pixel defining layer PDL. The light emitting layer EML may include an organic material layer. The organic material layer of the light emitting layer may include an organic light emitting layer, and may further include a hole injection/transport layer and/or an electron injection/transport layer.

The cathode electrode CAT may be disposed on the light emitting layer EML. The cathode electrode CAT may be a common electrode extended across all the pixels. The anode electrode ANO, the light emitting layer EML, and the cathode electrode CAT may constitute an organic light emitting element.

The thin film encapsulation layer EN including a first inorganic film EN**1**, a first organic film EN**2** and a second inorganic film EN**3** is disposed on the cathode electrode CAT. The first inorganic film EN**1** and the second inorganic film EN**3** may be in contact with each other at an end portion of the thin film encapsulation layer EN. The first organic film EN**2** may be sealed by the first inorganic film EN**1** and the second inorganic film EN**3**.

Each of the first inorganic film EN**1** and the second inorganic film EN**3** may include silicon nitride, silicon oxide, silicon oxynitride, or the like. The first organic film EN**2** may include an organic insulating material.

Hereinafter, an embodiment of an apparatus for manufacturing a display device used to form a polycrystalline silicon thin film included in the semiconductor layer ACT described above will be described.

FIG. **3**A is a perspective view illustrating an annealing process of a display device according to an embodiment. FIG. **3**B is a cross-sectional view illustrating an annealing process of a display device according to an embodiment. FIG. **4**A is a front view showing the shape of a final laser beam according to an embodiment. FIG. **4**B is a front view cumulatively illustrating the locations where the final laser beam is irradiated as the target substrate according to an embodiment moves.

Referring to FIGS. **3**A, **3**B, **4**A, and **4**B, in an annealing process of a display device according to an embodiment, the amorphous silicon thin film **11** may be disposed on the target substrate **10**. The target substrate **10** may be the base substrate SUB**1** described above with reference to FIG. **2**.

Herein, it is described as an example that the amorphous silicon thin film **11** is formed on the target substrate **10** and crystallized by the display device manufacturing apparatus **1000**, but the disclosure is not limited thereto. In an embodiment, for example, the amorphous silicon thin film **11** may be an amorphous semiconductor layer that further includes another material in addition to silicon.

The amorphous silicon thin film **11** may be formed by a method such as chemical vapor deposition (CVD) or plasma CVD. In an embodiment, the amorphous silicon thin film **11** may be formed using silicon or a silicon compound (e.g., SixGey).

The amorphous silicon thin film **11** may have a uniform thickness in each region, but is not limited thereto. In an embodiment, for example, as shown in FIG. **3**B, the amorphous silicon thin film **11** may include uneven top and bottom surfaces. In such an embodiment, the amorphous silicon thin film **11** may have different thicknesses in different regions due to its microstructure.

The display device manufacturing apparatus **1000** according to an embodiment may be a laser annealing apparatus that crystallizes the amorphous silicon thin film **11** into a polycrystalline silicon thin film **12** using (e.g., by radiating) a laser.

The display device manufacturing apparatus **1000** may provide a final laser beam FLB on the amorphous silicon thin film **11**. The display device manufacturing apparatus **1000** may be positioned on the target substrate **10** when providing the final laser beam FLB on the amorphous silicon thin film **11**. In an embodiment, for example, the display device manufacturing apparatus **1000** may be positioned above the target substrate **10** when providing the final laser beam FLB on the amorphous silicon thin film **11**.

The display device manufacturing apparatus **1000** may radiate the final laser beam FLB to the amorphous silicon thin film **11** located on the target substrate **10**. The amorphous silicon thin film **11** irradiated with the final laser beam FLB may be crystallized into the polycrystalline silicon thin film **12**.

A large amount of energy may be used for effective crystallization of the amorphous silicon thin film **11**. Therefore, it may be desirable for the energy provided per unit area of the region to be irradiated with the final laser beam FLB to be large. Accordingly, in an embodiment, the display device manufacturing apparatus **1000** may process a plurality of raw laser beams RLB (see FIG. **7**) to form the final laser beam FLB. The process of processing the raw laser beams RLB (see FIG. **7**) to form the final laser beam FLB will be described later with reference to FIG. **5** and the like.

When the amorphous silicon thin film **11** is crystallized at a temperature lower than the melting point, abnormal protrusions formed at grain boundaries can be reduced, thereby improving crystallinity. Therefore, it may be desirable to heat the amorphous silicon thin film **11** to a temperature lower than the melting point of the amorphous silicon thin film **11**. In an embodiment, for example, when the melting temperature of the amorphous silicon thin film **11** is about 1460° C., the final laser beam FLB may be controlled to heat the amorphous silicon thin film to a temperature of about 1300° C. to 1400° C. The final laser beam FLB may be irradiated to the amorphous silicon thin film **11** for several nanoseconds (ns) to tens of nanoseconds (ns).

The amorphous silicon contained in the amorphous silicon thin film **11** may be melted by a rapid increase in temperature due to the final laser beam FLB being radiated thereto, and then cooled and recrystallized. In this way, the amorphous silicon thin film **11** may be crystallized into the polycrystalline silicon thin film **12** by the final laser beam FLB. The repeated melting and recrystallization of the amorphous silicon thin film **11** may cause unevenness on the surface, which may increase the surface roughness. That is, the surface roughness of the polycrystalline silicon thin film **12** may be greater than that of the amorphous silicon thin film **11**.

The final laser beam FLB may be emitted in the form of a line beam extending in one direction and having a width in another direction. The final laser beam FLB may be emitted in a path direction Lz. The line shape of the final laser beam FLB may extend in a major axis direction Lx perpendicular to the path direction Lz, which is the emission direction, and may have a predetermined width in a minor axis direction Ly perpendicular to the path direction Lz.

In this specification and the drawings, the path direction Lz may refer to a travel direction of the laser, i.e., a direction along the movement path of the laser. The major axis direction Lx may refer to the extension direction of the laser, and the minor axis direction Ly may refer to the width direction of the laser. In some embodiments, the major axis direction Lx, the minor axis direction Ly, and the path direction Lz may be perpendicular to each other, but are not limited thereto. In the display device manufacturing apparatus **1000** according to an embodiment, the major axis direction Lx, the minor axis direction Ly, and the path direction Lz may be changed along the movement path of the laser.

As shown in FIGS. **3**B and **4**A, the final laser beam FLB may have a trapezoidal shape on a plane viewed in the major axis direction Lx. In an embodiment, for example, the final laser beam FLB may have a shape whose width increases from top to bottom on the plane viewed in the major axis direction Lx.

As shown in FIG. **4**A, the beam profile shape of the final laser beam FLB may be a flat top type. Accordingly, uniform energy may be transmitted to the silicon thin films **11** and **12** in the region where the laser is irradiated.

The major axis direction Lx, which is the extension direction of the final laser beam FLB, may be referred to as a major axis direction, and the length in the major axis direction Lx may be referred to as a major axis size Dx.

The larger the major axis size Dx is, the larger the area of the amorphous silicon thin film **11** crystallized is. In some embodiments, the major axis size Dx may be equal to the length of the target substrate **10** in the major axis direction Lx. Accordingly, the process may be completed with only one scan in one direction, thereby improving process efficiency. In an embodiment, the major axis size Dx may be in a range of about 1000 millimeters (mm) to about 2000 mm. In an embodiment, for example, the major axis size Dx may be about 1500 mm.

The minor axis direction Ly, which is the thickness direction perpendicular to the extension direction of the final laser beam FLB, may be referred to as a minor axis direction, and the length in the minor axis direction Ly may be referred to as a minor axis size Dy. In an embodiment, the minor axis size Dy of the final laser beam FLB may be in a range of about 95 micrometers (μm) to about 110 μm. By having the minor axis size Dy that is small, the display device manufacturing apparatus **1000** according to an embodiment may provide greater energy per unit area to the silicon thin films **11** and **12**, thereby improving crystallization quality.

As shown in FIG. **4**A, the minor axis size Dy may be a width at a point that is 90% of the total height from the lower edge of the final laser beam FLB. In an embodiment, for example, the minor axis size Dy may be a horizontal length in the minor axis direction Ly at a point that is 90% of the total height from the lower edge of the final laser beam FLB on a plane viewed in the major axis direction Lx.

In some embodiments, an inclined surface size S_Dy of the final laser beam FLB may be in a range of about 30 μm to about 45 μm. The inclined surface size S_Dy may be a width of a portion corresponding to 10% to 90% of the total height from the lower edge of the final laser beam FLB. In an embodiment, for example, the inclined surface size S_Dy may be a horizontal length in the minor axis direction Ly of the portion corresponding to 10% to 90% of the total height from the lower edge of the final laser beam FLB. The inclined surface size S_Dy may be measured with respect to one of both opposing side surfaces of the final laser beam FLB.

Although not shown in the drawings, the target substrate **10** on which the amorphous silicon thin film **11** is formed may be placed on a movable stage (not shown). While the final laser beam FLB is being radiated, the movable stage (not shown) may uniformly move the target substrate **10** in the direction of an arrow to ensure that the amorphous silicon thin film **11** on the target substrate **10** is evenly irradiated with the final laser beam FLB.

In another embodiment, for example, the display device manufacturing apparatus **1000** may move and irradiate the final laser beam FLB while the target substrate **10** on which the amorphous silicon thin film **11** is formed remains stationary. In still another embodiment, for example, the display device manufacturing apparatus **1000** may irradiate the final laser beam FLB while moving together with the target substrate **10** on which the amorphous silicon thin film **11** is formed. In such an embodiment, the display device manufacturing apparatus **1000** and the target substrate **10** on which the amorphous silicon thin film **11** is formed may move at a same moving speed, but they are not limited thereto and may move at different speeds.

In some embodiments, as shown in FIG. **4**B, the radiation region of the final laser beam FLB may overlap several times or more on the silicon thin films **11** and **12**. A scan interval I_Dy of the final laser beam FLB may be smaller than the minor axis size Dy. In an embodiment, the scan interval I_Dy may be about 2 μm. Since the non-overlap ratio between adjacent final laser beams FLB is a ratio of the scan interval I_Dy to the minor axis size Dy, the overlap ratio between adjacent final laser beams FLB may be in a range of about 97.5% to about 98.5%. Since the number of radiation times of the final laser beam FLB at one point of the silicon thin films **11** and **12** is determined by the ratio of the scan interval I_Dy to the minor axis size Dy, the number of radiation times of the final laser beam FLB at one point of the silicon thin films **11** and **12** may be in a range of 47 to 55 times.

In an embodiment, the energy density per unit area of the final laser beam FLB may be in a range of about 310 millijoules per square centimeter (mJ/cm^2) to about 320 mJ/cm^2.

The display device manufacturing apparatus **1000** according to an embodiment may have a high overlap ratio and a larger number of irradiation times to increase the amount of energy applied to the silicon thin films **11** and **12**, thereby improving crystallization quality. Accordingly, the amount of energy applied to the silicon thin films **11** and **12** may be increased without excessively increasing the output, oscillation frequency, or the like of the laser, thereby minimizing (or substantially reducing) thermal deformation and damage to optical components included in the display device manufacturing apparatus **1000**.

Even when any one of the target substrate **10** and the display device manufacturing apparatus **1000** moves, the crystallization process speed may be determined based on the relative speed between the target substrate **10** and the display device manufacturing apparatus **1000**. The process speed of the display device manufacturing apparatus **1000** according to an embodiment, i.e., the relative speed between the target substrate **10** and the display device manufacturing apparatus **1000**, may be about 20 millimeters per second (mm/s), but is not limited thereto. This process speed may be calculated by multiplying the frequency of the raw laser beam RLB (see FIG. **7**) by the scan interval I_Dy.

The top and bottom surfaces of the silicon thin films **11** and **12** may include unevenness. As shown in FIG. **3**B, the top surface of the silicon thin films **11** and **12** may have a highest point HP protruding most to the other side of the path direction Lz, and the bottom surface thereof may have a lowest point LP protruding most to one side of the path direction Lz.

In order to achieve uniform crystallization of the amorphous silicon thin film **11**, it is desired to provide uniform energy to each region. Substantially uniform energy may be provided to regions located within the depth of focus (DOF) of the final laser beam FLB from the focal plane. The depth of focus (DOF) refers to the distance where the focus is considered to be maintained, whether moving farther away from or closer to the focal plane. As the value of the depth of focus (DOF) for the final laser beam FLB increases, the region where substantially the same energy as the energy provided to the focal plane is provided may become broader. In other words, the focus of the final laser beam FLB may be formed within the amorphous silicon thin film **11**, and it may be desirable for the depth of focus (DOF) of the final laser beam FLB to be at least greater than a level difference d between the highest point HP and the lowest point LP in order to achieve uniform crystallization of the amorphous silicon thin film **11**.

FIG. **5** is a schematic block diagram showing an apparatus for manufacturing a display device according to an embodiment. FIG. **6** is a perspective view showing an apparatus for manufacturing a display device according to an embodiment.

Referring to FIGS. **5** and **6**, an embodiment of the display device manufacturing apparatus **1000** may be a solid-state laser annealing (SLA) apparatus. The solid-state laser annealing apparatus may have higher productivity, lower maintenance costs, and higher crystallization quality than an excimer laser annealing apparatus. The solid-state laser annealing apparatus refers to a laser device that uses a solid material as the medium for the laser, and the excimer laser annealing apparatus refers to a laser device that uses a gaseous material as the medium for the laser.

For example, unlike the excimer laser annealing apparatus, the solid-state laser annealing apparatus may be performed without gas filling, so that the display device manufacturing apparatus **1000** according to an embodiment may have high productivity and low maintenance costs due to no downtime for gas filling. The downtime refers to the time during which the crystallization process is not in progress.

In addition, the excimer laser annealing apparatus may be desired to operate in a state where the stage is tilted to prevent vertical mura, the scan interval is set to an integer multiple of the thin film transistor arrangement spacing to prevent diagonal mura that may occur as a result of tilting, and the stage is vector driven to minimize dead space caused by tilting. In an embodiment, the display device manufacturing apparatus **1000**, which is a solid-state laser annealing apparatus, may be performed without complex equipment operations such as tilting the stage, setting the scan interval to an integer multiple, and vector driving the stage, thereby achieving high process efficiency and high crystallization quality.

In some embodiments, the display device manufacturing apparatus **1000** may use at least one selected from ytterbium yag (Yb:YAG), neodymium yag (Nd:YAG), ruby (Cr:Al_2O_3), and titanium sapphire (Ti:Sapphire) as the medium for the laser.

The display device manufacturing apparatus **1000** may include a laser module **100**, a beam coupling module **200**, a beam mixing unit **300**, a micro smoother **400**, an optical module **500**, a seal box **600**, and a chamber **700**.

The laser module **100** may generate a plurality of raw laser beams RLB. The laser module **100** may include a plurality of laser generators. The laser module **100** may be a solid-state laser device that uses a solid material as a medium for the laser, as described above.

The beam coupling module **200** may be disposed at one side of the laser module **100**. In an embodiment, for example, the beam coupling module **200** may be disposed at one side of the laser module **100** in a second direction DR**2**. The beam coupling module **200** may convert the plurality of raw laser beams RLB generated from the laser module **100** into a plurality of coupling beams CPB. The beam coupling module **200** may organize the plurality of raw laser beams RLB into one or several groups to form the plurality of coupling beams CPB.

In the specification and the drawings, a first direction DR**1** and the second direction DR**2** cross each other as horizontal directions. For example, the first direction DR**1** and the second direction DR**2** may be orthogonal to each other. In addition, the third direction DR**3** crosses the first direction DR**1** and the second direction DR**2**, and may be, for example, perpendicular directions orthogonal to each other. In the specification, directions indicated by arrows of the first to third directions DR**1** to DR**3** in the drawings may be referred to as one side, and the opposite directions thereto may be referred to as the other side. However, if one side or the other side is not specified, it may not be limited to either one side or the other side. The first to third directions DR**1**, DR**2**, and DR**3** may be independent directions unrelated to the major axis direction Lx, the minor axis direction Ly, and the path direction Lz.

The beam mixing unit **300** may be disposed to one side of the beam coupling module **200**. In an embodiment, for example, the beam mixing unit **300** may be disposed to one side of the beam coupling module **200** in the second direction DR**2**. The beam mixing unit **300** may convert the plurality of coupling beams CPB into a plurality of mixing beams MXB. In an embodiment, the beam mixing unit **300** may form the plurality of mixing beams MXB by mixing the plurality of coupling beams CPB at a certain ratio. In such an embodiment, the plurality of mixing beams MXB may each include a same amount of the raw laser beams RLB.

The micro smoother **400** may be disposed to one side of the beam mixing unit **300**. In an embodiment, for example, the micro smoother **400** may be disposed at one side portion of the beam mixing unit **300** in the first direction DR**1**. The micro smoother **400** may be a device that produces regular or irregular wobbling of the laser beam. By intentionally causing the laser beam to wobble, the micro smoother **400** may effectively prevent horizontal mura that may occur during crystallization of the silicon thin film.

In an embodiment, for example, as described above with reference to FIG. **4**A and the like, the final laser beam FLB may have a flat top type profile shape. However, when some uneven stepped portions occur in the flat top shape or there are particles or the like in the path of the laser beam, energy imbalance may occur in a portion of the laser beam due to the stepped portions, the particles, or the like. Accordingly, when the portion of the laser beam in which such energy imbalance has occurred is used to scan the substrate while irradiating a certain area of the substrate, horizontal mura may occur along the portion in which the imbalance has occurred. In the display device manufacturing apparatus **1000** according to an embodiment, the micro smoother **400** may offset the energy imbalance by creating intentional wobble in the laser beam. Accordingly, the occurrence of horizontal mura may be effectively prevented.

The optical module **500** may be disposed on the beam mixing unit **300** and the micro smoother **400**. In an embodiment, for example, the optical module **500** may be disposed to one side of the beam mixing unit **300** and the micro smoother **400** in the third direction DR**3**. The optical module **500** may adjust the phase, travel distance, travel direction, and the like of the plurality of mixing beams MXB that have passed through the beam mixing unit **300**. The optical module **500** may determine the shape of the final laser beam FLB by adjusting the phase, travel distance, travel direction, or the like of the plurality of mixing beams MXB. The optical module **500** may include at least one selected from a phase retarder, a lens, and a mirror.

The seal box **600** may be disposed between the optical module **500** and the chamber **700**. In an embodiment, for example, the seal box **600** may be disposed between the optical module **500** and the chamber **700** in the third direction DR**3**. The seal box **600** may be a space for processing the final laser beam FLB before the final laser beam FLB is incident on the chamber **700**.

The chamber **700** may be disposed below the optical module **500**. In an embodiment, for example, the chamber **700** may be disposed at one side of the optical module **500** in a direction opposite to the third direction DR**3**. The chamber **700** may provide an internal space for performing an annealing process.

FIG. **7** is a perspective view showing a laser module according to an embodiment.

Referring to FIG. **7** in addition to FIGS. **5** and **6**, an embodiment of the laser module **100** may include a plurality of laser generators. In an embodiment, for example, the laser module **100** may include a first laser generator **101**, a second laser generator **102**, a third laser generator **103**, a fourth laser generator **104**, a fifth laser generator **105**, a sixth laser generator **106**, a seventh laser generator **107**, an eighth laser generator **108**, a ninth laser generator **109**, and a tenth laser generator **110**. In the drawing, an embodiment where the laser module **100** includes ten laser generators is shown, but the number of laser generators included in the laser module **100** is not limited thereto. The number of laser generators included in the laser module **100** may vary depending on the total output power, oscillation frequency, process speed, or the like of the laser module **100**.

The first to fifth laser generators **101** to **105** may be arranged side by side in the first direction DR**1**. The sixth to tenth laser generators **106** to **110** may be arranged side by side in the first direction DR**1**. The first laser generator **101** may be arranged side by side with the sixth laser generator **106** in the third direction DR**3**. The second laser generator **102** may be arranged side by side with the seventh laser generator **107** in the third direction DR**3**. The third laser generator **103** may be arranged side by side with the eighth laser generator **108** in the third direction DR**3**. The fourth laser generator **104** may be arranged side by side with the ninth laser generator **109** in the third direction DR**3**. The fifth laser generator **105** may be arranged side by side with the tenth laser generator **110** in the third direction DR**3**.

The laser module **100** may generate the raw laser beam RLB. In an embodiment, for example, the first laser generator **101** may generate a first raw laser beam RLB**1**, the second laser generator **102** may generate a second raw laser beam RLB**2**, the third laser generator **103** may generate a third raw laser beam RLB**3**, the fourth laser generator **104** may generate a fourth raw laser beam RLB**4**, the fifth laser generator **105** may generate a fifth raw laser beam RLB**5**, the sixth laser generator **106** may generate a sixth raw laser beam RLB**6**, the seventh laser generator **107** may generate a seventh raw laser beam RLB**7**, the eighth laser generator **108** may generate an eighth raw laser beam RLB**8**, the ninth laser generator **109** may generate a ninth raw laser beam RLB**9**, and the tenth laser generator **110** may generate a tenth raw laser beam RLB**10**.

The raw laser beam RLB may include at least one or more laser beams. In an embodiment, for example, the first to tenth raw laser beams RLB**1** to RLB**10** may each include two laser beams, but are not limited thereto.

The total output power of the laser module **100** may be in a range of about 5,760 watts (W) to about 9,000 watts (W). In an embodiment, for example, the total output power of the laser module **100** may be about 7200 W. The output power of each of the first to tenth laser generators **101** to **110** may be about 600 W to 900 W. Preferably, the output power of each of the first to tenth laser generators **101** to **110** may be about 720 W. The output power of each of the first to tenth laser generators **101** to **110** may be the same as each other, but the disclosure is not limited thereto. By increasing the number of laser generators and setting the output power of each laser generator to be about 1000 W or less, the display device manufacturing apparatus **1000** according to an embodiment may effectively prevent thermal deformation due to high energy accumulation in various optical components included in the display device manufacturing apparatus **1000**, thereby increasing the lifespan.

The oscillation frequency of the laser module **100**, e.g., the oscillation frequency of each of the first to tenth laser generators **101** to **110**, may be in a range of about 5,000 hertz (Hz) to about 15,000 hertz (Hz). In an embodiment, for example, the oscillation frequency of each of the first to tenth laser generators **101** to **110** may be about 10,000 hertz (Hz). The display device manufacturing apparatus **1000** according to an embodiment may improve the efficiency and yield of the annealing process by using a laser with a high repetition rate. In such an embodiment, since the oscillation frequency of the laser module **100** is about 15,000 hertz (Hz) or less, thermal deformation due to high energy accumulation in various optical components included in the display device manufacturing apparatus **1000** may be effectively prevented, thereby increasing the lifespan.

The raw laser beam RLB may be a laser beam in the ultraviolet region. For example, the wavelength of the raw laser beam RLB may be in a range of about 337 nanometers (nm) to about 357 nm. In an embodiment, for example, the wavelength of the raw laser beam RLB may be in a range of about 341 nm to about 345 nm. The display device manufacturing apparatus **1000** according to an embodiment may have high transmittance by using the raw laser beam RLB in the ultraviolet region having a high wavelength. As a result, the depth of focus (DOF) may increase and thus uniform crystallization may be achieved.

FIG. **8** is a cross-sectional view showing a beam coupling module according to an embodiment. FIG. **9** is a perspective view illustrating a group coupling unit and a pulse extender module according to an embodiment. FIG. **10** is a schematic diagram showing paths of laser beams passing through a beam coupling module and a beam mixing unit according to an embodiment.

Referring to FIGS. **8** to **10** in addition to FIGS. **5** and **6**, the beam coupling module **200** may include a plurality of coupling module devices **201** to **210**, at least one scraper unit **220**, a group telescope unit **230**, and a group coupling unit **240**.

The beam coupling module **200** may include the plurality of coupling module devices. In an embodiment, for example, the beam coupling module **200** may include a first beam coupling module **201**, a second beam coupling module **202**, a third beam coupling module **203**, a fourth beam coupling module **204**, a fifth beam coupling module **205**, a sixth beam coupling module **206**, a seventh beam coupling module **207**, an eighth beam coupling module **208**, a ninth beam coupling module **209**, and a tenth beam coupling module **210**. In the drawing, an embodiment where the beam coupling module **200** includes ten beam coupling module devices is shown, but the number of the beam coupling modules **200** is not limited thereto. The number of the beam coupling modules **200** may vary depending on the number of the laser generators of the laser module **100**.

The first to fifth beam coupling modules **201** to **205** may be arranged side by side in the first direction DR**1**. The sixth to tenth beam coupling modules **206** to **210** may be arranged side by side in the first direction DR**1**. The first beam coupling module **201** may be arranged side by side with the sixth beam coupling module **206** in the third direction DR**3**. The second beam coupling module **202** may be arranged side by side with the seventh beam coupling module **207** in the third direction DR**3**. The third beam coupling module **203** may be arranged side by side with the eighth beam coupling module **208** in the third direction DR**3**. The fourth beam coupling module **204** may be arranged side by side with the ninth beam coupling module **209** in the third direction DR**3**. The fifth beam coupling module **205** may be arranged side by side with the tenth beam coupling module **210** in the third direction DR**3**.

The first to tenth beam coupling modules **201** to **210** may be disposed at one side of the first to tenth laser generators **101** to **110** in the second direction DR**2**, respectively. The first to tenth beam coupling modules **201** to **210** may receive the first to tenth raw laser beams RLB**1** to RLB**10** from the first to tenth laser generators **101** to **110**, respectively.

The first to tenth beam coupling modules **201** to **210** may each include at least one shutter **201***a* to **210***a*. In an embodiment, for example, the first beam coupling module **201** may include a first shutter **201***a*, the second beam coupling module **202** may include a second shutter **202***a*, the third beam coupling module **203** may include a third shutter **203***a*, the fourth beam coupling module **204** may include a fourth shutter **204***a*, the fifth beam coupling module **205** may include a fifth shutter **205***a*, the sixth beam coupling module **206** may include a sixth shutter **206***a*, the seventh beam coupling module **207** may include a seventh shutter **207***a*, the eighth beam coupling module **208** may include an eighth shutter **208***a*, the ninth beam coupling module **209** may include a ninth shutter **209***a*, and the tenth beam coupling module **210** may include a tenth shutter **210***a*.

The first to tenth shutters **201***a* to **210***a* may adjust the incidence of the first to tenth raw laser beams RLB**1** to RLB**10** into the first to tenth beam coupling modules **201** to **210**, respectively. The shutter operation speed of the first to tenth shutters **201***a* to **210***a* may be in a range of about 20 milliseconds (ms) to about 500 milliseconds (ms). In an embodiment, for example, the shutter operation speed of the first to tenth shutters **201***a* to **210***a* may be in a range of about 20 milliseconds (ms) to about 100 milliseconds (ms).

The first to tenth beam coupling modules **201** to **210** may each include at least one high-speed photodiode **201***b* to **210***b*, at least one low-speed photodiode **201***c* to **210***c*, and at least one single beam monitor **201***d* to **210***d*. In an embodiment, for example, the first beam coupling module **201** may include a first high-speed photodiode **201***b*, a first low-speed photodiode **201***c*, and a first single beam monitor **201***d*. The second beam coupling module **202** may include a second high-speed photodiode **202***b*, a second low-speed photodiode **202***c*, and a second single beam monitor **202***d*. The third beam coupling module **203** may include a third high-speed photodiode **203***b*, a third low-speed photodiode **203***c*, and a third single beam monitor **203***d*. The fourth beam coupling module **204** may include a fourth high-speed photodiode **204***b*, a fourth low-speed photodiode **204***c*, and a fourth single beam monitor **204***d*. The fifth beam coupling module **205** may include a fifth high-speed photodiode **205***b*, a fifth low-speed photodiode **205***c*, and a fifth single beam monitor **205***d*. The sixth beam coupling module **206** may include a sixth high-speed photodiode **206***b*, a sixth low-speed photodiode **206***c*, and a sixth single beam monitor **206***d*. The seventh beam coupling module **207** may include a seventh high-speed photodiode **207***b*, a seventh low-speed photodiode **207***c*, and a seventh single beam monitor **207***d*. The eighth beam coupling module **208** may include an eighth high-speed photodiode **208***b*, an eighth low-speed photodiode **208***c*, and an eighth single beam monitor **208***d*. The ninth beam coupling module **209** may include a ninth high-speed photodiode **209***b*, a ninth low-speed photodiode **209***c*, and a ninth single beam monitor **209***d*. The tenth beam coupling module **210** may include a tenth high-speed photodiode **210***b*, a tenth low-speed photodiode **210***c*, and a tenth single beam monitor **210***d*.

The high-speed photodiodes **201***b* to **210***b*, the low-speed photodiodes **201***c* to **210***c*, and the single beam monitors **201***d* to **210***d* may be configured to monitor the state of the raw laser beam RLB. In an embodiment, for example, the high-speed photodiodes **201***b* to **210***b* may monitor the pulse shape of the raw laser beam RLB, the low-speed photodiodes **201***c* to **210***c* may monitor the peak power of the raw laser beam RLB, and the single beam monitors **201***d* to **210***d* may monitor other characteristics, such as pulse duration and wavelength, of the raw laser beam RLB.

The scraper unit **220** may group each of the first to tenth raw laser beams RLB provided from the first to tenth beam coupling modules **201** to **210** to form the coupling beam CPB. The scraper unit **220** may include a first scraper unit **221**, a second scraper unit **222**, and a third scraper unit **223**.

The first scraper unit **221** may be connected to the first beam coupling module **201** and the sixth beam coupling module **206**. The second scraper unit **222** may be connected to the second beam coupling module **202**, the third beam coupling module **203**, the seventh beam coupling module **207**, and the eighth beam coupling module **208**. The third scraper unit **223** may be connected to the fourth beam coupling module **204**, the fifth beam coupling module **205**, the ninth beam coupling module **209**, and the tenth beam coupling module **210**.

The first scraper unit **221** may use the first raw laser beam RLB**1** to generate a first coupling beam CPB**1**. The first scraper unit **221** may use the sixth raw laser beam RLB**6** to generate a second coupling beam CPB**2**. The second scraper unit **222** may use the second raw laser beam RLB**2**, the third raw laser beam RLB**3**, the seventh raw laser beam RLB**7**, and the eighth raw laser beam RLB**8** to generate a fourth coupling beam CPB**4** and a fifth coupling beam CPB**5**. The third scraper unit **223** may use the fourth raw laser beam RLB**4**, the fifth raw laser beam RLB**5**, the ninth raw laser beam RLB**9**, and the tenth raw laser beam RLB**10** to generate a third coupling beam CPB**3** and a sixth coupling beam CPB**6**.

The second scraper unit **222** may include a first pre-mixer **222***a*, and the third scraper unit **223** may include a second pre-mixer **223***a*. The first and second pre-mixers **222***a* and **223***a* may be optical devices that transmit and reflect incident laser beams at a 1:1 ratio. The second scraper unit **222** and the third scraper unit **223** may generate the coupling beams CPB by grouping the plurality of raw laser beams RLB using the first pre-mixer **222***a* and the second pre-mixer **223***a*, respectively.

The group telescope unit **230** may adjust the degree of diffusion of the coupling beam CPB incident on the group telescope unit **230**. In an embodiment, for example, the group telescope unit **230** may adjust the degree of diffusion of each laser beam included in the coupling beam CPB to provide the coupling beam CPB to the group coupling unit **240**.

The telescope unit **230** may include a first telescope unit **231**, a second telescope unit **232**, and a third telescope unit **233**. The first telescope unit **231** may connect the first scraper unit **221** to the group coupling unit **240**. The second telescope unit **232** may connect the second scraper unit **222** to the group coupling unit **240**. The third telescope unit **233** may connect the third scraper unit **223** to the group coupling unit **240**.

The group coupling unit **240** may provide the coupling beam CPB to the beam mixing unit **300**. In an embodiment, for example, the group coupling unit **240** may change the path of the coupling beam CPB traveling in the first direction DR1 to the second direction DR2. The coupling beam CPB whose path has been changed by the group coupling unit **240** may move to the beam mixing unit **300**.

In some embodiments, the beam coupling module **200** may further include a pulse extender module (an optical pulse duration extender module (OPDEM)) **250**.

The pulse extender module **250** may be disposed between the group coupling unit **240** and the beam mixing unit **300**. The pulse extender module **250** may extend the pulse duration of the final laser beam FLB.

In an embodiment, for example, the pulse extender module **250** may adjust the time at which each coupling beam CPB is incident on the beam mixing unit **300** by controlling the movement path and travel direction of the coupling beam CPB. In one example, the pulse extender module **250** may include first to sixth beam path adjusters **251**, **252**, **253**, **254**, **255**, and **256** that adjust the movement paths and travel directions of the first to sixth coupling beams CPB**1** to CPB**6**, respectively. The first to sixth beam path adjusters **251**, **252**, **253**, **254**, **255**, and **256** may include at least one selected from a phase retarder, a lens, and a mirror.

The pulse shape and pulse duration of the final laser beam FLB may be determined by the combination of the mixing beams MXB (see FIG. **11**). Since the mixing beam MXB is generated using the coupling beam CPB, the pulse duration of the final laser beam FLB may be extended by adjusting the time at which each coupling beam CPB is incident on the beam mixing unit **300**.

For example, if the time for the first coupling beam CPB**1** to depart from the group coupling unit **240**, pass through the first beam path adjuster **251**, and arrive at the beam mixing unit **300** is shorter than the time for the second coupling beam CPB**2** to depart from the group coupling unit **240**, pass through the second beam path adjuster **252**, and arrive at the beam mixing unit **300**, the first coupling beam CPB**1** and the second coupling beam CPB**2** may form the final laser beam FLB having lower energy and a longer pulse duration through mutual reinforcement/cancellation interference. If the arrival times of the first coupling beam CPB**1** and the second coupling beam CPB**2** are the same as each other, the final laser beam FLB having higher energy may be formed without increasing the pulse duration.

The display device manufacturing apparatus **1000** according to an embodiment may freely or efficiently adjust the pulse duration and energy intensity of the final laser beam FLB by including the pulse extender module **250**. Accordingly, various pulse shapes of the final laser beam FLB may be implemented.

In an embodiment, the pulse duration of the final laser beam FLB by the pulse extender module **250** may be in a range of about 14 nanoseconds (ns) to about 20 nanoseconds (ns).

FIG. **11** is a schematic diagram showing the path of laser beams passing through a beam mixing unit according to an embodiment. FIG. **12** is a schematic diagram showing a process in which each coupling beam is distributed into a mixing beam in a beam mixing unit according to an embodiment.

Referring to FIGS. **11** and **12** in addition to FIGS. **5**, **6**, **8**, and **10**, an embodiment of the beam mixing unit **300** may mix the plurality of coupling beams CPB to generate the plurality of mixing beams MXB. In an embodiment, for example, the beam mixing unit **300** may mix the plurality of coupling beams CPB at various ratios to generate first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6**.

The first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6** may have substantially the same energy intensity as each other. The display device manufacturing apparatus **1000** according to an embodiment may generate the first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6** of the same energy using the beam mixing unit **300** to form the final laser beam FLB having uniform energy. Accordingly, it is possible to effectively prevent the vertical mura phenomenon that may occur in the silicon thin film during the annealing process.

The beam mixing unit **300** may include a reflection mirror SPL**10**, a first splitter SPL**11**, and a second splitter SPL**21**. The beam mixing unit **300** may generate the first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6** having the same energy by adjusting the energy and paths of the coupling beams CPB incident on the reflection mirror SPL**10**, the first splitter SPL**11**, and the second splitter SPL**21**.

The reflection mirror SPL**10** may change the path of the coupling beam CPB without changing energy by reflecting the incident coupling beam CPB.

The first splitter SPL**11** may transmit and reflect the incident coupling beam CPB at a ratio of 1:1. The ratio of the energy intensity of the transmitted coupling beam CPB and the reflected coupling beam CPB at the first splitter SPL**11** may be 1:1.

The second splitter SPL**21** may transmit and reflect the incident coupling beam CPB at a ratio of 2:1. The ratio of the energy intensity of the transmitted coupling beam CPB and the reflected coupling beam CPB at the second splitter SPL**21** may be 2:1.

In an embodiment, for example, when the initial energy intensity of each of the first to tenth raw laser beams RLB**1** to RLB**10** is 100, the first coupling beam CPB**1** and the second coupling beam CPB**2** may be generated using the first raw laser beam RLB**1** and the sixth raw laser beam RLB**6**, respectively, so that the energy intensity of the first coupling beam CPB**1** and the second coupling beam CPB**2** may each be **100**. Since the third coupling beam CPB**3** and the sixth coupling beam CPB**6** are generated using the fourth raw laser beam RLB**4**, the fifth raw laser beam RLB**5**, the ninth raw laser beam RLB**9**, and the tenth raw laser beam RLB**10**, the energy intensity of the third coupling beam CPB**3** and the sixth coupling beam CPB**6** may each be **200**. Since the fourth coupling beam CPB**4** and the fifth coupling beam CPB**5** are generated using the second raw laser beam RLB**2**, the third raw laser beam RLB**3**, the seventh raw laser beam RLB**7**, and the eighth raw laser beam RLB**8**, the energy intensity of the fourth coupling beam CPB**4** and the fifth coupling beam CPB**5** may each be **200**.

First, as shown in first schematic diagram G1 of FIG. **12**, the first coupling beam CPB**1** having an energy of **100** may be split into two laser beams each having an energy of **50** through the first splitter SPL**11** in first position. The laser beams each having an energy of **50** may each be split into two laser beams each having an energy of **25** and two laser beams each having an energy of **25'** through the first splitter SPL**11** in second position.

In this specification and drawings, the distinction between the energy of **25** and the energy of **25'** is made merely for simplicity of description, but they refer to energy of the same intensity. In the following, an energy of **100** and an energy of **100'** are the same.

The laser beam having an energy of **25** may be split into a laser beam having an energy of **25** (**4/6**) and a laser beam having an energy of **25** (**2/6**) through the second splitter SPL**21**. The laser beam having an energy of **25** (**2/6**) may be split into two laser beams each having an energy of **25** (**1/6**) through the first splitter SPL**11** in third position.

The laser beam having an energy of **25'** may be split into two laser beams each having an energy of **25** (**3/6**)' through the first splitter SPL**11** in third position.

The laser beam having an energy of **25** (**4/6**) may be included in the first mixing beam MXB**1** and the fourth mixing beam MXB**4**. The laser beam having an energy of **25** (**1/6**) and the laser beam having an energy of **25** (**3/6**)' may be combined with each other and included in the second mixing beam MXB2, the third mixing beam MXB3, the fifth mixing beam MXB5, and the sixth mixing beam MXB6.

As a result, each of the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may be equally provided with an energy of **25** (**4**/**6**) from the first raw laser beam RLB1.

Next, as shown in second schematic diagram G2 of FIG. **12**, the second coupling beam CPB2 having an energy of **100** may also be divided in the same manner as the first coupling beam CPB1 to form the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6.

As a result, each of the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may be equally provided with an energy of **25** (**4**/**6**) from the sixth raw laser beam RLB6.

Next, as shown in third schematic diagram G3 of FIG. **12**, the third coupling beam CPB3 having an energy of **200** may be split into a laser beam having an energy of **100** and a laser beam having an energy of **100'** through the first splitter SPL11 in first position.

The laser beam having an energy of **100** may be split into a laser beam having an energy of **100** (**4**/**6**) and a laser beam having an energy of **100** (**2**/**6**) through the second splitter SPL21. The laser beam having an energy of **100** (**2**/**6**) may be split into two laser beams each having an energy of **100** (**1**/**6**) through the first splitter SPL11 in second position.

The laser beam having an energy of **100'** may be divided into two laser beams each having an energy of **100** (**3**/**6**)' through the first splitter SPL11 in second position.

The laser beam having an energy of **100** (**4**/**6**) may be included in the first mixing beam MXB1. The laser beam having an energy of **100** (**1**/**6**) and the laser beam having an energy of **100** (**3**/**6**)' may be combined with each other and included in the second mixing beam MXB2 and the third mixing beam MXB3.

Next, as shown in sixth schematic diagram G6 of FIG. **12**, the sixth coupling beam CPB6 having an energy of **200** may be split into a laser beam having an energy of **200** (**1**/**3**) and a laser beam having an energy of **200** (**2**/**3**) through the second splitter SPL21.

The laser beam having an energy of **200** (**2**/**3**) may be split into two laser beams each having an energy of **200** (**1**/**3**) through the first splitter SPL11.

The laser beam having an energy of **200 1**/**3** may be included in the fourth mixing beam MXB4, the fifth mixing beam MXB5, and the sixth mixing beam MXB6.

As a result, each of the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may be equally provided with an energy of **100** (**4**/**6**) from the fourth raw laser beam RLB4, the fifth raw laser beam RLB5, the ninth raw laser beam RLB9, and the tenth raw laser beam RLB10. That is, the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may each be provided with an energy of **25** (**4**/**6**) from the fourth raw laser beam RLB4, an energy of **25** (**4**/**6**) from the fifth raw laser beam RLB5, an energy of **25** (**4**/**6**) from the ninth raw laser beam RLB9, and an energy of **25** (**4**/**6**) from the tenth raw laser beam RLB10.

Next, as shown in fourth schematic diagram G4 of FIG. **12**, the fourth coupling beam CPB4 having an energy of **200** may be split into a laser beam having an energy of **100** and a laser beam having an energy of **100'** through the first splitter SPL11 in first position.

The laser beam having an energy of **100** may be split into a laser beam having an energy of **100** (**4**/**6**) and a laser beam having an energy of **100** (**2**/**6**) through the second splitter SPL21. The laser beam having an energy of **100** (**2**/**6**) may be split into two laser beams each having an energy of **100** (**1**/**6**) through the first splitter SPL11 in second position.

The laser beam having an energy of **100'** may be split into two laser beams each having an energy of **100** (**3**/**6**)' through the first splitter SPL11 in second position.

The laser beam having an energy of **100** (**4**/**6**) may be included in the fourth mixing beam MXB4. The laser beam having an energy of **100** (**1**/**6**) and the laser beam having an energy of **100** (**3**/**6**)' may be combined with each other and included in the fifth mixing beam MXB5 and the sixth mixing beam MXB6.

Next, as shown in fifth schematic diagram G5 of FIG. **12**, the fifth coupling beam CPB5 having an energy of **200** may be split into a laser beam having an energy of **200** (**1**/**3**) and a laser beam having an energy of **200** (**2**/**3**) through the second splitter SPL21.

The laser beam having an energy of **200** (**2**/**3**) may be split into two laser beams each having an energy of **200** (**1**/**3**) through the first splitter SPL11.

The laser beam having an energy of **200** (**1**/**3**) may be included in the first mixing beam MXB1, the second mixing beam MXB2, and the third mixing beam MXB3.

As a result, each of the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may be equally provided with an energy of **100** (**4**/**6**) from the second raw laser beam RLB2, the third raw laser beam RLB3, the seventh raw laser beam RLB7, and the eighth raw laser beam RLB8. In such an embodiment, as described above, each of the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may be provided with an energy of **25** (**4**/**6**) from the second raw laser beam RLB2, an energy of **25** (**4**/**6**) from the third raw laser beam RLB3, an energy of **25** (**4**/**6**) from the seventh raw laser beam RLB7, and an energy of **25** (**4**/**6**) from the eighth raw laser beam RLB8.

In such an embodiment, as shown in first to sixth schematic diagrams G1 to G6, the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may each receive an energy of **25** (**4**/**6**) equally from the first to tenth raw laser beams RLB1 to RLB10. Accordingly, the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 may have substantially the same energy intensity. The display device manufacturing apparatus **1000** according to an embodiment may generate the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 having a same energy using the beam mixing unit **300** to form the final laser beam FLB having uniform energy. Accordingly, it is possible to effectively prevent the vertical mura phenomenon that may occur in the silicon thin film during the annealing process.

In the display device manufacturing apparatus **1000** according to thane embodiment, it is desired for the first to sixth mixing beams MXB1, MXB2, MXB3, MXB4, MXB5, and MXB6 to have a same intensity of energy, and also to uniformly include the first to tenth raw laser beams RLB1 to RLB10 by the same amount of energy.

FIG. **13** is a cross-sectional view illustrating a beam mixing unit, a micro smoother, an optical module, a seal box, a chamber, and a basic frame according to an embodiment. FIG. **14** is a perspective view illustrating the beam mixing unit, the micro smoother, and the optical module according to an embodiment.

Referring to FIGS. **13** and **14** in addition to FIGS. **5** and **6**, in an embodiment, the mixing beam MXB generated in the beam mixing unit **300** may reach the target substrate **10** through the micro smoother **400**, the optical module **500**, the seal box **600**, and the chamber **700**.

The micro smoother **400** may be disposed on one side of the beam mixing unit **300**. In an embodiment, for example, the micro smoother **400** may be disposed on one side of the beam mixing unit **300** in the first direction DR**1**. The micro smoother **400** may include a first micro smoother **410** and a second micro smoother **420**. The micro smoother **400** will be described later with reference to FIGS. **15** and **16**.

The optical module **500** may be disposed on the beam mixing unit **300** and the micro smoother **400**. In an embodiment, for example, the optical module **500** may be disposed on one side of the beam mixing unit **300** and the micro smoother **400** in the third direction DR**3**. The optical module **500** may include at least one selected from at least one phase retarder, a lens, and a mirror. The optical module **500** may adjust the movement path and traveling direction of the laser beam. The optical module **500** may provide the laser beam provided from the micro smoother **400** to the chamber **700**.

The optical module **500** may include a first optical module **510**, a second optical module **520**, a third optical module **530**, a fourth optical module **540**, and a fifth optical module **550**.

The first optical module **510** may be disposed on the micro smoother **400**. The second optical module **520** may be disposed on the beam mixing unit **300**. The fourth optical module **540** and the fifth optical module **550** may be disposed on the chamber **700**. The third optical module **530** may be disposed between the second optical module **520** and the fourth optical module **540**.

The first to third optical modules **510**, **520**, and **530** may be disposed in a first sector SECT**1**. The fourth and fifth optical modules **540** and **550** may be disposed in a second sector SECT**2**. The second sector SECT**2** may be an area in which the chamber **700** is disposed, and the first sector SECT**1** may be an area disposed on one side of the second sector SECT**2** and may be an area in which the chamber **700** is not disposed.

In some embodiments, the third optical module **530** and the fourth optical module **540** may be disposed to be spaced apart from each other. in an embodiment, for example, the third optical module **530** and the fourth optical module **540** may be disposed to be spaced apart from each other by a predetermined distance D_**500** in the first direction DR**1**. In another embodiment, all the first to fifth optical modules **510**, **520**, **530**, **540**, and **550** may be disposed to be spaced apart.

In the display device manufacturing apparatus **1000** according to an embodiment, the third optical module **530** and the fourth optical module **540** may be disposed to be spaced apart from each other, such that the transfer of vibration occurring in the first sector SECT**1** to the second sector SECT**2** may be substantially reduced or minimized.

In some embodiments, although not illustrated in the drawings, the display device manufacturing apparatus **1000** may further include a wave plate.

The wave plate may be disposed between the beam mixing unit **300** and the optical module **500**. In an embodiment, for example, the wave plate may be disposed between the beam mixing unit **300** and the first micro smoother **410**, between the first micro smoother **410** and the second micro smoother **420**, or between the second micro smoother **420** and the optical module **500**.

In some embodiments, the wave plates may be disposed in paths of first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6**, respectively. In such embodiments, the number of wave plates may be 6, but is not limited thereto, and the number of wave plates may be less or more than the number of first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6**.

The wave plate may adjust the degree of polarization of the final laser beam FLB by adjusting the degree of polarization of the mixing beam MXB. In an embodiment, for example, the wave plate may cause at least one selected from the first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6** to be S-polarized (vertically polarized) and the rest to be P-polarized (horizontally polarized). In an embodiment, for example, the wave plate may cause three of the first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6** to be S-polarized and the remaining three thereof to be P-polarized.

By including the wave plate, the display device manufacturing apparatus **1000** according to an embodiment may evenly include an S-polarized beam and a P-polarized beam in the final laser beam FLB. Accordingly, the extension directions of the annealed silicon thin film particles are evenly distributed in the major axis direction Lx and the minor axis direction Ly, such that the reliability of the silicon thin film may be improved.

The seal box **600** may be disposed below the optical module **500**. The chamber **700** may be disposed below the seal box **600**. The seal box **600** and chamber **700** will be described later with reference to FIGS. **17** to **20**.

The display device manufacturing apparatus **1000** according to an embodiment may further include a basic frame **800**.

The basic frame **800** may include a first stage frame **810**, a second stage frame **820**, a third stage frame **830**, and a fourth stage frame **840**.

The first stage frame **810** may include a plurality of pedestals. The plurality of pedestals of the first stage frame **810** may be disposed to be spaced apart along the first direction DR**1**. However, the disclosure is not limited thereto, and the plurality of pedestals of the first stage frame **810** may be disposed to be spaced apart along the second direction DR**2**. Since the first stage frame **810** includes the plurality of pedestals spaced apart from each other, vibration transfer between parts may be substantially reduced or minimized.

The second stage frame **820** may be disposed on the first stage frame **810**. The second stage frame **820** may include an integrally formed pedestal, that is, a pedestal integrally formed as a single unitary and indivisible part. Since the first stage frame **810** includes an integrally formed pedestal, the load of the equipment disposed on top may be distributed.

The third stage frame **830** may be disposed on the second stage frame **820**. The third stage frame **830** may include a plurality of pedestals. The plurality of pedestals of the third stage frame **830** may be disposed to be spaced apart along the first direction DR**1**. However, the disclosure is not limited thereto, and the plurality of pedestals of the third stage frame **830** may be disposed to be spaced apart along the second direction DR**2**. Since the third stage frame **830** includes the plurality of pedestals spaced apart from each other, the vibration transfer between parts may be substantially reduced or minimized.

In some embodiments, although not illustrated in the drawings, the extension direction of the plurality of pedestals of the first stage frame **810** and the extension direction of the plurality of pedestals of the third stage frame **830** may be different from each other. In an embodiment, for example, the extension direction of the plurality of pedestals of the first stage frame **810** may be in the first direction DR**1**, and the extension direction of the plurality of pedestals of the third stage frame **830** may be in the second direction DR**2**. In another embodiment, for example, the extension direction of the plurality of pedestals of the first stage frame **810** may be in the second direction DR**2**, and the extension direction of the plurality of pedestals of the third stage frame **830** may be in the first direction DR**1**. In the basic frame **800** according to an embodiment, the extension direction of the plurality of pedestals of the first stage frame **810** and the extension direction of the plurality of pedestals of the third stage frame **830** may be disposed differently from each other, such that the load of the equipment disposed on top may be distributed.

The fourth stage frame **840** may be disposed on the third stage frame **830**. The fourth stage frame **840** may include a first pedestal **841** and a second pedestal **842**. The first pedestal **841** may be disposed in the first sector SECT**1**, and the second pedestal **842** may be disposed in the second sector SECT**2**.

The first pedestal **841** may support the beam mixing unit **300**, the micro smoother **400**, the first optical module **510**, the second optical module **520**, and the third optical module **530**. In an embodiment, for example, the first pedestal **841** may overlap the beam mixing unit **300**, the micro smoother **400**, the first optical module **510**, the second optical module **520**, and the third optical module **530** in the third direction DR**3**.

The second pedestal **842** may support the fourth optical module **540**, the fifth optical module **550**, the seal box **600**, and the chamber **700**. In an embodiment, for example, the second pedestal **842** may overlap the fourth optical module **540**, the fifth optical module **550**, the seal box **600**, and the chamber **700** in the third direction DR**3**.

The first pedestal **841** and the second pedestal **842** may be disposed to be spaced apart from each other. In an embodiment, for example, the first pedestal **841** and the second pedestal **842** may be disposed to be spaced apart from each other by a predetermined distance D_**800** in the first direction DR**1**.

In the display device manufacturing apparatus **1000** according to an embodiment, the first pedestal **841** and the second pedestal **842** may be disposed to be spaced apart from each other, such that the transfer of vibration occurring in the equipment, vibration occurring around the equipment, or the like may be substantially reduced or minimized.

FIG. **15** is a cross-sectional view illustrating a micro smoother according to an embodiment. FIG. **16** is a perspective view illustrating an operation method of a micro smoother according to an embodiment.

Referring to FIGS. **15** and **16** in addition to FIGS. **5**, **6**, **13**, and **14**, an embodiment of the micro smoother **400** may include the first micro smoother **410** and the second micro smoother **420**. The first micro smoother **410** and the second micro smoother **420** may be disposed to be spaced apart from each other along the path direction Lz of the mixing beam MXB.

Each of the first micro smoother **410** and the second micro smoother **420** may include a smoother cover **401**, a floating box **402**, an elastic tube **403**, a smoother optical module **404**, a weight **405**, and a smoother driver **406**.

The smoother cover **401** may provide a space in which the floating box **402**, the elastic tube **403**, and the smoother optical module **404** are disposed. Nitrogen (N_2) gas may be filled in the space partitioned by the smoother cover **401**. Accordingly, it is possible to effectively prevent external impacts from being transmitted to the smoother optical module **404** inside the floating box **402** or vibrations inside the micro smoother **400** from being transmitted to other parts of the display device manufacturing apparatus **1000**.

The floating box **402** may be disposed in a space partitioned (or enclosed) by the smoother cover **401**. The floating box **402** may float within a space partitioned by the smoother cover **401**. In an embodiment, for example, floating box **402** may be an air bearing box or an air floating box. As the floating box **402** floats, it is possible to effectively prevent external impacts from being transmitted to the smoother optical module **404** inside the floating box **402** or vibrations inside the micro smoother **400** from being transmitted to other parts of the display device manufacturing apparatus **1000**.

The elastic tube **403** may be connected to the inner surface of the smoother cover **401** and the outer surface of the floating box **402**. In FIG. **15**, for convenience of illustration, four elastic tubes **403** are shown, but the number of elastic tubes **403** is not limited thereto. The elastic tube **403** may effectively prevent the floating box **402** that floats from moving excessively.

The smoother optical module **404** may adjust the movement path and traveling direction of the mixing beam MXB. In an embodiment, for example, the smoother optical module **404** may include a small and lightweight optical device. In an embodiment, the smoother optical module **404** included in the first micro smoother **410** may be a telescope lens. In an embodiment, the smoother optical module **404** included in the second micro smoother **420** may be a homogenizer. The telescope lens of the first micro smoother **410** and the homogenizer of the second micro smoother **420** may adjust the movement path and traveling direction in the major axis direction Lx but are not limited thereto, and may also adjust the movement path and traveling direction of the minor axis direction Ly.

In some embodiments, all the first micro smoother **410** and the second micro smoother **420** may be driven, or only one of the first micro smoother **410** and the second micro smoother **420** may be driven.

A small and lightweight optical device included in the smoother optical module **404** may correspond to each of a plurality of mixing beams MXB. In an embodiment, for example, a total of six small and lightweight optical devices may be disposed corresponding to the first to sixth mixing beams MXB**1**, MXB**2**, MXB**3**, MXB**4**, MXB**5**, and MXB**6**, respectively. The number of small and lightweight optical devices may be variously modified depending on the number of mixing beams MXB.

The weight **405** may support the smoother cover **401**. The weight **405** may prevent the smoother cover **401** from shaking due to external impacts.

The smoother driver **406** may provide driving force to the floating box **402**. The smoother driver **406** may provide driving force to move the smoother optical module **404** in the major axis direction Lx of the laser beam. In an embodiment, for example, the smoother driver **406** may provide driving force to move the floating box **402** along the second direction DR**2**. However, the disclosure is not limited thereto, and the smoother driver **406** may provide driving force such that the floating box **402** moves along the first direction DR**1**.

The micro smoother **400** may be a device that generates shaking of the laser beam regularly or irregularly. By intentionally causing the laser beam to shake by the micro smoother **400**, horizontal mura that may occur during crystallization of the silicon thin film may be effectively prevented.

In an embodiment, for example, as described above with reference to FIG. 4A, the final laser beam FLB may have a flat top type profile shape. However, when some uneven stepped portions occur in the flat top shape or there are particles or the like in the path of the laser beam, energy imbalance may occur in a portion of the laser beam due to the stepped portions, the particles, or the like. In this way, when a certain area of the substrate is irradiated with the portion of the laser beam in which such energy imbalance has occurred to perform scanning, horizontal mura may occur along the portion in which the imbalance has occurred.

In the display device manufacturing apparatus **1000** according to an embodiment, the first micro smoother **410** and the second micro smoother **420** may be driven in the major axis direction Lx. In an embodiment, for example, the first micro smoother **410** and the second micro smoother **420** may perform regular or irregular vibration movements in the major axis direction Lx. Accordingly, a central hit point CH of the final laser beam FLB may deviate from a center line CL of the target substrate **10** by a first vibration distance D_**400** in the major axis direction Lx. In an embodiment, the first vibration distance D_**400** may be in a range of about 30 mm to about 60 mm. In an embodiment, for example, the first vibration distance D_**400** may be in a range of about 40 mm to about 50 mm.

In this way, by causing the micro smoother **400** to intentionally shake the laser beam, it is possible to effectively prevent a specific hit point from continuously moving along the minor axis direction Ly, that is, the scan direction, and forming horizontal mura, and the energy imbalance may be offset. Accordingly, the occurrence of horizontal mura may be effectively prevented.

In an embodiment, as described above, the laser beam of the display device manufacturing apparatus **1000** may have a high repetition rate. Accordingly, the number of times the final laser beam FLB is radiated to the target substrate **10** may increase. Accordingly, it is desired to drive the micro smoother **400** at high speed according to the number of radiation times of the final laser beam FLB. The micro smoother **400** according to an embodiment may realize high-speed driving by including a small and lightweight optical device. In an embodiment, the frequency of the smoother driver **406** may be in a range of about 50 hertz (Hz) to about 250 hertz (Hz).

In some embodiments, the target substrate **10** may be driven directly in the major axis direction Lx. Accordingly, the central hit point CH of the final laser beam FLB may deviate from the center line CL of the target substrate **10** by the first vibration distance D_**400** in the major axis direction Lx. In an embodiment, the driving speed of the target substrate **10** may be the same as the scanning speed of the target substrate **10**. In an embodiment, for example, the driving speed of the target substrate **10** may be about 20 millimeters per second (mm/s). However, the disclosure is not limited thereto, and the driving speed of the target substrate **10** may be faster than the scanning speed of the target substrate **10**. In this case, the driving speed of the target substrate **10** may be about 30 mm/s.

FIG. **17** is a cross-sectional view illustrating an optical module, a seal box, and a chamber according to an embodiment. FIG. **18** is a cross-sectional view illustrating an operation method of a beam cutter according to an embodiment. FIG. **19** is a plan view illustrating a fixing chuck and a target substrate according to an embodiment. FIG. **20** is a perspective view illustrating a profile meter, a power meter, and a measurement aperture according to an embodiment.

Referring to FIGS. **17** to **20** in addition to FIGS. **5**, **6**, and **13**, the laser beam that has passed through the optical module **500** may pass further through the seal box **600** and reach the chamber **700**.

The seal box **600** may provide a space for processing the final laser beam FLB that has passed through the optical module **500**. In an embodiment, the seal box **600** may include a seal box window **610**, a beam cutter **620**, and a beam dump **630**.

The seal box window **610** may be disposed between the optical module **500** and the seal box **600**. In an embodiment, for example, the seal box window **610** may be disposed on a partition wall positioned between the optical module **500** and the seal box **600**. The seal box window **610** may be an optical passage through which the final laser beam FLB that has passed through the optical module **500** may move inside the seal box **600**. The seal box window **610** may include or be made of a material that allows the final laser beam FLB to pass therethrough.

The beam cutter **620** may adjust the major axis size Dx of the final laser beam FLB. The beam cutter **620** may overlap at least a portion of the final laser beam FLB in the path direction Lz of the laser beam. The beam cutter **620** may reflect at least a portion of the final laser beam FLB and transmit the remaining portion thereof.

In an embodiment, for example, the beam cutter **620** may include a first portion **621** and a second portion **622** disposed to be spaced apart from each other in the major axis direction Lx of the laser beam, and a third portion **623** disposed between the first portion **621** and the second portion **622**.

The first portion **621** and the second portion **622** of the beam cutter **620** may be reflection areas, and the third portion **623** may be a transmission area. In an embodiment, the first portion **621** and the second portion **622** may include a reflection mirror, and the third portion **623** may include a transmission window. In another embodiment, the first portion **621** and the second portion **622** may be treated with a high reflection (HR) coating (or include a high reflection coating film), and the third portion **623** may be treated with an anti-reflection (AR) coating (or include an anti-reflection (AR) coating film).

The first portion **621** of the beam cutter **620** may overlap one end FLBla of a before-processing final laser beam FLB**1** in the path direction Lz. The second portion **622** of the beam cutter **620** may overlap the other end FLB**1***b* of the before-processing final laser beam FLB**1** in the path direction Lz. The third portion **623** of the beam cutter **620** may overlap a central portion FLB**1***c* positioned between the one end FLBla and the other end FLB**1***b* of the before-processing final laser beam FLB**1** in the path direction Lz.

The one end FLBla and the other end FLB**1***b* of the before-processing final laser beam FLB**1** may be reflected from the first portion **621** and the second portion **622** of the beam cutter **620**, respectively, and the central portion FLBlc of the before-processing final laser beam FLB**1** may be transmitted through the third portion **623**. The transmitted central portion FLB**1***c* may form a final laser beam FLB**2** processed by adjusting the major axis size Dx. The processed final laser beam FLB**2** may reach the target substrate **10**.

In some embodiments, the beam cutter **620** may include only one of the first portion **621** and the second portion **622**. In such embodiments, the beam cutter **620** may reflect only one portion of the one end FLBla and the other end FLB**1***b* of the final laser beam FLB to adjust the major axis size Dx of the final laser beam FLB.

In an embodiment, the reflection surfaces of the first portion **621** and the second portion **622** of the beam cutter **620** may include a heat-resistant material. In an embodiment, for example, the first portion **621** and the second portion **622** of the beam cutter **620** may include glass. Since the display device manufacturing apparatus **1000** according to an embodiment uses a laser beam having a high repetition rate and high energy, the reflection surfaces of the first portion **621** and the second portion **622** may continuously receive energy from the laser before the temperature cools down despite the cooling system. Since the reflection surfaces of the first portion **621** and the second portion **622** include a heat-resistant material, thermal deformation due to temperature rise may be effectively prevented.

The beam dump **630** may attenuate or extinguish the energy of the laser beam reflected from the beam cutter **620** and the laser beam reflected from a measurement aperture **770**. The beam dump **630** may include a reflection mirror on the inner surface of the beam dump **630**. The beam dump **630** may include a cooling system, such as coolant, inside the beam dump **630**.

In an embodiment, for example, the one end FLB**1***a* and the other end FLB**1***b* of the before-processing final laser beam FLB**1** reflected from the first portion **621** and the second portion **622** of the beam cutter **620** may move inside the beam dump **630**. The laser beam that enters the beam dump **630** may be continuously reflected by the reflection mirror inside the beam dump **630** and may collide with the cooling system. Accordingly, the energy of the laser beam may be attenuated or extinguished.

In an embodiment, the chamber **700** may include a chamber window **710**, a stage **720**, a fixing chuck **730**, a stage driver **740**, a profile meter **750**, a power meter **760**, and a measurement aperture **770**.

The chamber window **710** may be disposed between the seal box **600** and the chamber **700**. In an embodiment, for example, the chamber window **710** may be disposed on a partition wall positioned between the seal box **600** and the chamber **700**. The chamber window **710** may be an optical passage through which the final laser beam FLB that has passed through the seal box **600** may move inside the chamber **700**. The chamber window **710** may include or be made of a material that allows the final laser beam FLB to pass therethrough.

The stage **720** may be disposed inside the chamber **700**. The stage **720** may provide a space in which the target substrate **10** is seated. In an embodiment, for example, the target substrate **10** may be seated on the stage **720** during an annealing process.

The fixing chuck **730** may be disposed on the stage **720**. The fixing chuck **730** may fix the target substrate **10** to the stage **720** such that the target substrate **10** is not separated from the stage **720** during the annealing process. In an embodiment, the fixing chuck **730** may be a vacuum chuck using suction force. In another embodiment, the fixing chuck **730** may be an electrostatic chuck using electromagnetic force. However, the disclosure is not limited thereto, and the fixing chuck **730** may include any device capable of fixing the target substrate **10** to the stage **720**.

In some embodiments, as illustrated in FIG. **19**, the fixing chuck **730** may be larger than the target substrate **10**. The target substrate **10** may include a laser irradiation area LSA and a laser non-irradiation area NLSA. The laser irradiation area LSA may be an area where the final laser beam FLB is irradiated, and the laser non-irradiation area NLSA may be an area where the final laser beam FLB is not irradiated.

In some embodiments, the fixing chuck **730** may not apply fixing force to the target substrate **10** in the area overlapping the laser irradiation area LSA, and may apply fixing force to the target substrate **10** in the area overlapping the laser non-irradiation area NLSA.

In an embodiment, for example, depending on the roughness of the surface of the fixing chuck **730**, on which the fixing chuck **730** and the target substrate **10** are in contact, cone mura may occur by a height difference due to the roughness. The fixing chuck **730** according to an embodiment may effectively prevent cone mura due to surface roughness of the fixing chuck **730** by not applying fixing force in the area overlapping the laser irradiation area LSA.

The stage driver **740** may provide driving force to the stage **720**. In an embodiment, for example, the stage driver **740** may move the stage **720** in the first direction DR**1** or the minor axis direction Ly. In another embodiment, for example, the stage driver **740** may move the stage **720** in the second direction DR**2** or the major axis direction Lx. In another embodiment, for example, the stage driver **740** may move the stage **720** in the third direction DR**3** or the path direction Lz.

The profile meter **750** may measure the energy profile of the final laser beam FLB. In an embodiment, for example, the profile meter **750** may measure the pulse shape of the final laser beam FLB reaching the target substrate **10**.

The power meter **760** may measure the energy power of the final laser beam FLB. In an embodiment, for example, the power meter **760** may measure the energy density of the final laser beam FLB reaching the target substrate **10**. In such an embodiment, at least one selected from the profile meter **750** and the power meter **760** may be individually or collectively referred to as a meter.

The measurement aperture **770** may adjust the shape and size of the final laser beam FLB reaching the profile meter **750** and the power meter **760**. When the final laser beam FLB provided to the target substrate **10** is directly radiated to the profile meter **750** and the power meter **760**, the profile meter **750**, the power meter **760**, and the equipment of the vicinity may be damaged by the high energy of the final laser beam FLB. The measurement aperture **770** according to an embodiment may effectively prevent such damage by adjusting the shape and size of the final laser beam FLB.

As illustrated in FIG. **20**, the measurement aperture **770** may overlap the final laser beam FLB**2** processed by passing through the beam cutter **620** in the path direction Lz. The measurement aperture **770** may reflect at least a portion of the processed final laser beam FLB**2** and transmit the remaining portion.

In an embodiment, for example, the measurement aperture **770** may include a reflection area **770***a* and a transmission area **770***b*. The transmission area **770***b* may overlap a beam receiving portion (not illustrated) of the profile meter **750** and a beam receiving portion (not illustrated) of the power meter **760** in the path direction Lz. In some embodiments, the reflection area **770***a* may surround the transmission area **770***b*, but is not limited thereto.

In an embodiment, the reflection area **770***a* may include a reflection mirror, and the transmission area **770***b* may include a hole or a transmission window. In another embodiment, the reflection area **770***a* may be treated with a high reflection (HR) coating, and the transmission area **770***b* may be treated with an anti-reflection (AR) coating.

The reflection area **770***a* may overlap one portion FLB**2***a* of the processed final laser beam FLB**2** in the path direction Lz. The transmission area **770***b* may overlap the other portion FLB**2***b* of the processed final laser beam FLB**2** in the path direction Lz.

The one portion FLB**2***a* of the processed final laser beam FLB**2** may be reflected in the reflection area **770***a*, and the other portion FLB**2***b* of the processed final laser beam FLB**2** may be transmitted in the transmission area **770***b*. The shape and size of the transmitted other portion FLB2*b* may be adjusted to form a reduced final laser beam FLB3. The reduced final laser beam FLB3 may reach the beam receiving portion (not illustrated) of the profile meter **750** and the beam receiving portion (not illustrated) of the power meter **760**.

The reflected one portion FLB2*a* may move inside the beam dump **630**, and the energy thereof may be attenuated or extinguished in the same manner as the laser beam reflected from the beam cutter **620**.

By including the measurement aperture **770**, the display device manufacturing apparatus **1000** according to an embodiment may effectively prevent the final laser beam FLB from reaching other portions except the beam receiving portion (not illustrated) of the profile meter **750** and the beam receiving portion (not illustrated) of the power meter **760**, and thus may minimize equipment damage.

The display device manufacturing apparatus **1000** according to an embodiment is the solid-state laser annealing apparatus, which uses a laser with higher energy and higher repetition rate than the excimer laser annealing apparatus, and has a faster process speed than the excimer laser annealing apparatus, so that it is desired that optical system configurations and other equipment designs suitable for such characteristics are included.

FIG. **21** is a microscopic image illustrating particles of a polycrystalline silicon thin film according to a conventional embodiment. FIG. **22** is a microscopic image illustrating particles of a polycrystalline silicon thin film according to an embodiment. FIG. **23** is a photograph illustrating a macro inspection of a polycrystalline silicon thin film according to the conventional embodiment. FIG. **24** is a photograph illustrating a macro inspection of a polycrystalline silicon thin film according to an embodiment.

Referring to FIGS. **21** to **24**, FIGS. **21** and **23** are photographs of the polycrystalline silicon thin film **12** that has been annealed using an excimer laser annealing apparatus as the display device manufacturing apparatus **1000** according to the conventional embodiment. FIGS. **22** and **24** are photographs of the polycrystalline silicon thin film **12** that has been annealed using a solid-state laser annealing apparatus as the display device manufacturing apparatus **1000** according to an embodiment.

As illustrated in FIG. **21**, the average of a first particle size R**1** of the polycrystalline silicon thin film **12** manufactured using the display device manufacturing apparatus **1000** according to the conventional embodiment may be approximately within 310 nm. On the other hand, as illustrated in FIG. **22**, the average of a second particle size R**2** of the polycrystalline silicon thin film **12** manufactured using the display device manufacturing apparatus **1000** according to an embodiment may be approximately within 340 nm.

The display device manufacturing apparatus **1000** according to the conventional embodiment is an excimer laser annealing apparatus, and since the length of the laser wavelength is in a range of about 300 nm to about 315 nm, the average of the first particle size R**1** may be about 310 nm or less. On the other hand, the display device manufacturing apparatus **1000** according to an embodiment is a solid-state laser annealing apparatus, and since the laser wavelength is in a range of about 337 nm to about 357 nm, the average of the first particle size R**1** may be about 340 nm or less.

In addition, the standard deviation of the second particle size R**2** may be smaller than the standard deviation of the first particle size R**1** of the polycrystalline silicon thin film **12** manufactured using the display device manufacturing apparatus **1000** according to the conventional embodiment. The display device manufacturing apparatus **1000** according to an embodiment may have higher energy, a higher repetition rate, and a faster process speed than the display device manufacturing apparatus **1000** according to the conventional embodiment, so that the particle size distribution may be more even.

As illustrated in FIG. **23**, the polycrystalline silicon thin film **12** manufactured using the display device manufacturing apparatus **1000** according to the conventional embodiment may include vertical mura. On the other hand, as illustrated in FIG. **24**, the polycrystalline silicon thin film **12** manufactured using the display device manufacturing apparatus **1000** according to an embodiment may not include vertical mura.

In the display device manufacturing apparatus **1000** according to an embodiment, vertical mura may not occur due to characteristics such as a shorter scan interval, higher irradiation count, higher energy, and higher frequency than the display device manufacturing apparatus **1000** according to the conventional embodiment.

In addition, the display device manufacturing apparatus **1000** according to an embodiment, which is a solid-state laser annealing apparatus, may operate without performing a complex equipment operation such as stage tilting, a scan interval that is an integer multiple of the thin film transistor arrangement spacing, and stage vector driving used in an excimer laser annealing apparatus to effectively prevent vertical mura, such that process efficiency and high crystallization quality may be achieved.

The invention should not be construed as being limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete and will fully convey the concept of the invention to those skilled in the art.

While the invention has been particularly shown and described with reference to embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit or scope of the invention as defined by the following claims.

What is claimed is:

1. An apparatus for manufacturing a display device, the apparatus comprising:

a chamber;

a laser module comprising a plurality of laser generators which respectively generates a plurality of raw laser beams using a solid material as a medium;

a beam coupling module which combines the plurality of raw laser beams into a plurality of coupling beams;

a beam mixing unit which converts the plurality of coupling beams into a plurality of mixing beams;

a micro smoother which induces vibration movements of the plurality of mixing beams;

an optical module which converts the plurality of mixing beams passed through the micro smoother into a final laser beam, and directs the final laser beam to the chamber; and

a seal box disposed between the optical module and the chamber.

2. The apparatus of claim **1**, wherein

a major axis size of the final laser beam is in a range of about 1,000 millimeters to about 2,000 millimeters,

a minor axis size of the final laser beam is in a range of about 95 micrometers to about 110 micrometers, and

a horizontal length of a lateral inclined surface of the final laser beam is in a range of about 30 micrometers to about 45 micrometers.

3. The apparatus of claim 1, wherein
a total laser output of the laser module is in a range of about 5,760 watts to about 9,000 watts,
an output of each of the plurality of laser generators is in a range of about 600 watts to about 900 watts, and
an oscillation frequency of each of the plurality of laser generators is in a range of about 5,000 hertz to about 15,000 hertz.

4. The apparatus of claim 1, wherein
each of the plurality of raw laser beams is a laser beam in an ultraviolet region, and
a wavelength of each of the plurality of raw laser beams is in a range of about 337 nanometers to about 357 nanometers.

5. The apparatus of claim 1, wherein
the beam coupling module comprises:
a shutter which receives the plurality of raw laser beams; and
a photodiode and a beam monitor which monitor the plurality of raw laser beams,
wherein an operation speed of the shutter is in a range of about 20 milliseconds to about 100 milliseconds.

6. The apparatus of claim 1, wherein the beam coupling module comprises:
a scraper unit which generates the plurality of coupling beams, wherein the scraper unit groups the plurality of raw laser beams to be included in the plurality of coupling beams;
a group telescope unit which adjusts a degree of diffusion of the plurality of coupling beams; and
a group coupling unit which provides the plurality of coupling beams to the beam mixing unit.

7. The apparatus of claim 6, wherein
the scraper unit comprises a pre-mixer, and
the pre-mixer transmits and reflects the plurality of coupling beams which are incident at a 1 ratio of 1:1.

8. The apparatus of claim 6, wherein the beam coupling module further comprises a pulse extender module disposed between the group coupling unit and the beam mixing unit, wherein the pulse extender module extends a pulse duration of the final laser beam.

9. The apparatus of claim 1, wherein the beam mixing unit comprises:
a reflection mirror which reflects the plurality of incident coupling beams incident thereto;
a first splitter which transmits and reflects the plurality of incident coupling beams at a ratio of 1:1; and
a second splitter which transmits and reflects the plurality of incident coupling beams at a ratio of 2:1.

10. The apparatus of claim 9, wherein the beam mixing unit generates the plurality of mixing beams by combining the plurality of coupling beams at a same ratio.

11. The apparatus of claim 1, wherein the micro smoother provides a driving force to regularly or irregularly vibrate a hit point of the final laser beam with respect to a direction perpendicular to a scan direction of the final laser beam.

12. The apparatus of claim 11, wherein the micro smoother comprises at least one selected from:
a first micro smoother comprising a telescope lens; and
a second micro smoother comprising a homogenizer.

13. The apparatus of claim 11, wherein the micro smoother comprises:
a cover;
a floating box disposed in an inner space of the cover in a floating state; and
an elastic tube connecting the floating box to the cover.

14. The apparatus of claim 1, further comprising:
a first section where the micro smoother is located;
a second section where the chamber is located; and
a basic frame disposed below the chamber and the micro smoother, wherein
the basic frame comprises a first pedestal disposed in the first section and a second pedestal disposed in the second section, and
the first pedestal and the second pedestal are disposed to be spaced apart from each other.

15. The apparatus of claim 14, wherein
the optical module comprises a first sub-optical module disposed in the first section, and a second sub-optical module disposed in the second section, and
the first sub-optical module and the second sub-optical module are disposed to be spaced apart from each other.

16. The apparatus of claim 1, wherein the seal box comprises:
a beam cutter which reflects a portion of the final laser beam and transmits a remaining portion of the final laser beam; and
a beam dump which attenuates or extinguishes energy of the portion of the final laser beam reflected from the beam cutter.

17. The apparatus of claim 16, wherein
the beam cutter comprises a reflection area which reflects the final laser beam, and
a surface of the reflection area includes glass.

18. The apparatus of claim 16, further comprising:
a reflection mirror disposed on an inner surface of the beam dump, and
a cooler disposed inside the beam dump.

19. The apparatus of claim 1, wherein the chamber comprises:
a meter which measures a profile or power of the final laser beam; and
a measurement aperture disposed on the meter, wherein the measurement aperture reflects a portion of the final laser beam and transmits a remaining portion.

20. The apparatus of claim 19, wherein
the measurement aperture comprises:
a reflection area which reflects the final laser beam; and
a transmission area which transmits the final laser beam,
the reflection area comprises a high reflection coating film, and
the transmission area comprises an anti-reflection coating film or a hole.

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